Positive, monotone, and convex Gaussian process regression with Laplacian eigenfunctions

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Abstract

We develop a common construction for imposing positivity, monotonicity, or convexity throughout an interval in Hilbert space Gaussian process regression. The model represents the process as a finite sum of basis functions with Gaussian coefficients whose variances are given by the kernel's spectral density. We retain this representation and express each restriction as a threshold on one coefficient. Integrating out this coefficient gives a strongly log-concave marginal posterior with a normal survival weight, and conditional truncated normal sampling recovers the full constrained law. We characterize the thresholds through polynomial equations and bound posterior error from conservative numerical evaluation in total variation. Under spectral regularity, these posteriors converge in quadratic Wasserstein distance to the corresponding underlying posteriors as the computational interval expands and higher frequencies are retained, controlling uniform discrepancies in the response and derivatives through the constrained order. Mercury lidar and leaf light-response applications illustrate the consequences for derivative inference.

Keywords: Bayesian regression, continuum constraint, derivative inference, Gaussian process, Hilbert space approximation

1 Introduction

Scientific information about a response often concerns its sign or how it changes over an interval. Positivity restricts the response itself, while monotonicity and convexity restrict its slope and curvature when the corresponding derivatives exist. These properties need not be visible in noisy observations, so a Gaussian process (GP) that describes the data closely can still assign probability to curves that conflict with the proposed scientific model. We therefore seek a posterior law that imposes the restriction throughout a specified interval while retaining joint uncertainty about the response and its derivatives.

The Hilbert space GP (HSGP) approximation of Solin and Särkkä (2020) represents the process by a finite expansion in Laplacian eigenfunctions on an interval extending beyond the region of interest. This extension, called padding, is used to reduce boundary effects, while the rank, or number of retained functions, controls the finest variation represented for a given computational interval (Riutort-Mayol et al., 2023). The basis functions are independent of the covariance kernel and its parameters. The kernel enters through spectral values that determine Gaussian coefficient variances, allowing the same basis and its analytic derivatives to be reused across covariance specifications. Shape restrictions, however, concern the entire curve obtained by summing the oscillatory basis functions. Their validity is not determined by the signs of the individual coefficients, and checking a finite set of locations can miss a violation between them. The question addressed here is how

to condition this spectral model on the complete interval restriction while retaining its basis and kernel-derived Gaussian coefficient law.

Our construction begins with positivity. A random intercept shifts the whole curve above zero when it exceeds the negative minimum of the spectral part. The same argument extends to monotonicity and convexity whenever the retained span contains a direction with a uniformly positive selected derivative, because a sufficiently large coefficient along that direction makes the full curve satisfy the restriction for any fixed values of the remaining coefficients. Thus the continuum event becomes a lower bound on one Gaussian coefficient, with a threshold determined by the remaining coefficients. Under Gaussian observations, integrating out that coefficient gives an explicit marginal posterior weighted by a normal survival probability. A subsequent draw from its truncated conditional normal restores the full conditioned law. This representation makes continuum conditioning available within the HSGP expansion without changing the probability model through a nonlinear transformation of the curve.

1.1 Related work and contributions

Global derivative restrictions and derivative inference have precedents in Bayesian regression, where piecewise-linear isotonic models permit flat and increasing segments (Neelon and Dunson, 2004), while smooth parameterizations and regression splines give other monotone priors (Shively et al., 2009). Ordered Bernstein coefficients combine global monotonicity with variable selection or clustering (Curtis and Ghosh, 2011; Wilson et al., 2020), with the aerosol analysis of Wilson et al. (2020) illustrating derivative inference through the derivative of a monotone differential-pressure curve. For convex regression, maxima of random affine functions address multivariate inputs and permit nonsmooth surfaces (Hannah and Dunson, 2011), for which support and posterior consistency results are available. Integrating squared GPs provides smooth shape restrictions (Lenk and Choi, 2017), with spectral constructions also available for grouped data (Lenk et al., 2024). These approaches establish that global shape information and derivative inference can be incorporated in Bayesian regression in several ways. They also make the choice of probability model substantive: transforming or integrating a GP generally gives a different law from conditioning a specified Gaussian model on a shape event.

The closest constructions condition finite Gaussian expansions using bases chosen to express global shape restrictions as coefficient inequalities (Maatouk and Bay, 2017). These constraints hold throughout the domain, and subsequent work develops general linear inequalities, noisy observations, multiple constraints, and smooth bases (López-Lopera et al., 2018, 2020; Maatouk et al., 2025a). Within this line of work, Ray et al. (2020) combine smooth relaxations of coefficient constraints with elliptical slice sampling and circulant embedding, while Gaussian simulation on correlated subdomains addresses larger problems (Maatouk et al., 2025b). Our construction shares the aim of conditioning a finite Gaussian model on a global restriction. Its distinctive task is to characterize that restriction in the Laplacian–Dirichlet spectral space, whose oscillatory basis requires a different description of the constraint.

Other GP approaches impose derivative signs at virtual locations (Riihimäki and Vehtari, 2010; Agrell, 2019), use randomized optimization for constrained inference (Zhang et al., 2026), or project GP curves onto the monotone-function space (Lin and Dunson, 2014). Virtual-location restrictions provide a finite description of the constraint, whereas projection defines the fitted curve by a separate map. Here the target is the Gaussian regression posterior conditioned on the complete continuum event in the retained spectral space. The scalar threshold describes that event exactly, so its numerical evaluation can be studied separately from the choice of conditioning target.

The first contribution is a common representation for positivity, monotonicity, and convexity, together with explicit directions and sufficient rank and padding conditions for each case. Sign reversal gives the corresponding nonpositive, nonincreasing, and concave models. The representation also applies to joint restrictions whenever the retained span contains a direction with the required signs at all selected derivative orders. We give conditions on the HSGP direction coefficients and threshold reductions covering all 26 combinations without opposing constraints.

To evaluate the thresholds for the individual constraints, a cosine transformation reduces the search for stationary locations to polynomial equations, with endpoints, repeated roots, and constant cases included. These characterizations make the thresholds amenable to global evaluation. The resulting marginal posterior is strongly log-concave in whitened coordinates, and a total variation bound quantifies the change in its reconstructed law when a conservative numerical threshold is used. The computational cost is a global maximization at each target evaluation, so the representation offers an alternative to bases with direct coefficient inequalities, rather than a general efficiency advantage.

The second contribution connects finite continuum conditioning to conditioning the underlying GP. Covariance approximation alone does not settle this question because the indicator of a sign event is discontinuous at curves that touch its boundary. For each individual constraint, we establish a null boundary and control function-space moments, obtaining quadratic Wasserstein convergence of the conditioned posterior as padding and frequency cutoff grow. Under the stated spectral regularity, convergence holds in the uniform norm for the response and all derivatives up to the constrained order. It therefore also gives uniform convergence of their posterior means and covariance functions. The result includes Matérn smoothness greater than zero for positivity, greater than one for monotonicity, and greater than two for convexity.

Two applications show why retaining joint uncertainty about the response and its derivatives matters. Under the differential absorption lidar approximation, mercury concentration is proportional to the range derivative of a latent negative log signal ratio, so nonnegative concentration requires that curve to be nondecreasing. For leaf light-response data, concavity instead encodes a nonincreasing marginal gain in assimilation as irradiance increases, without requiring assimilation itself to increase throughout the range. These analyses illustrate how a scientifically specified shape restriction changes derivative inference.

This paper is organized as follows. Section 2 introduces the spectral model and its conditioning events, followed by the Gaussian coefficient construction for positivity, monotonicity, convexity, and their joint combinations in Section 2.1. Section 3 studies posterior geometry, numerical conditioning error, and convergence to the conditioned underlying GP, and Section 4 describes posterior computation. Section 5 presents the mercury lidar and leaf light-response applications, followed by the discussion in Section 6. All proofs are given in the supplementary material.

Stan implementations of the monotone and convex models, together with R code for reproducing the two applications, are available from the author's GitHub repository.

2 Methods

The Hilbert space Gaussian process (HSGP) construction of Solin and Särkkä (2020) approximates a stationary GP by a finite expansion in Laplacian eigenfunctions with Gaussian coefficients. The domain determines the basis functions, while the covariance kernel determines the coefficient variances. Changing the kernel or its parameters therefore leaves the basis and its derivatives unchanged, allowing their values at observation and prediction locations to be computed once and reused (Riutort-Mayol et al., 2023). For shape-constrained regression, a further advantage is that

the response, slope, and curvature are linear functions of the same coefficients. We can therefore express all three restrictions in a common coefficient space and obtain their conditioned laws from the ordinary Gaussian regression posterior.

Let $k(u)$ be a real stationary covariance function with a spectral density, where $u = x - x'$ is the separation between inputs. We write $S_k : \mathbb{R} \to [0, \infty)$ for its nonnegative, even, integrable spectral density and use the Fourier convention

$$k(u) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{\mathrm{i}\omega u} S_k(\omega)\, d\omega, \qquad u \in \mathbb{R}.$$

Here ω is angular frequency, and $S_k(\omega)$ is the spectral density associated with the covariance function k through this Fourier relation, evaluated at frequency ω. The HSGP expansion uses this density to assign variances at a discrete set of frequencies determined by the Laplacian on a bounded interval.

Fix a center $c \in \mathbb{R}$, a half-width $L > 0$, an integer $m \geq 1$, and a compact interval $I = [x_-, x_+]$ of positive length inside $(c - L, c + L)$. On the padded interval $[c - L, c + L]$, the negative Laplacian with zero boundary values has eigenvalues ω_j^2 and orthonormal eigenfunctions $\phi_{j,L}$. Using the spectral density S_k, these functions define the finite Gaussian expansion

$$\begin{aligned} \phi_{j,L}(x) &= L^{-1/2} \sin(\omega_j(x - c + L)), \quad \omega_j = j\pi/(2L), \\ f_{\boldsymbol{a}}(x) &= a_0 + \sum_{j=1}^{m} a_j \phi_{j,L}(x), \qquad\qquad a_0 \sim N(\mu_0, \sigma_0^2), \quad a_j \sim N(0, S_k(\omega_j)). \end{aligned} \tag{1}$$

The coefficients are independent, with $\sigma_0 > 0$ and $S_k(\omega_j) > 0$ at each retained frequency, so the finite coefficient law is nondegenerate.

Under this prior, the covariance is

$$\mathrm{Cov}(f_{\boldsymbol{a}}(x), f_{\boldsymbol{a}}(x')) = \sigma_0^2 + \sum_{j=1}^{m} S_k(\omega_j) \phi_{j,L}(x) \phi_{j,L}(x').$$

The sum approximates the stationary covariance $k(x - x')$, while σ_0^2 is the contribution of the random intercept. Because the eigenfunctions vanish at the padded endpoints, the finite covariance is generally nonstationary. Defining the basis on an interval larger than I places this artificial boundary outside the region of inference without requiring the response to vanish at the endpoints of I. The intercept permits uncertainty in the overall response level without changing slope or curvature.

For observations $\boldsymbol{y} \in \mathbb{R}^n$ at $x_i \in I$, the Gaussian observation model is

$$\boldsymbol{Y} \mid \boldsymbol{a} \sim N((f_{\boldsymbol{a}}(x_1), \ldots, f_{\boldsymbol{a}}(x_n))^T, \boldsymbol{\Sigma}),$$

where the noise covariance $\boldsymbol{\Sigma}$ is positive definite and the errors are independent of the coefficients. For given covariance and noise parameters and a chosen basis, we study the coefficient posterior conditional on the observed responses and their locations.

Write $\boldsymbol{a} = (a_0, \ldots, a_m)^T$. If $\boldsymbol{X}$ has rows $(1, \phi_{1,L}(x_i), \ldots, \phi_{m,L}(x_i))$, set $\boldsymbol{m}_0 = (\mu_0, 0, \ldots, 0)^T$ and $\boldsymbol{V}_0 = \mathrm{diag}(\sigma_0^2, S_k(\omega_1), \ldots, S_k(\omega_m))$. The coefficients and observations then have the joint Gaussian distribution

$$\begin{pmatrix} \boldsymbol{a} \\ \boldsymbol{Y} \end{pmatrix} \sim N\left(\begin{pmatrix} \boldsymbol{m}_0 \\ \boldsymbol{X}\boldsymbol{m}_0 \end{pmatrix}, \begin{pmatrix} \boldsymbol{V}_0 & \boldsymbol{V}_0 \boldsymbol{X}^T \\ \boldsymbol{X}\boldsymbol{V}_0 & \boldsymbol{X}\boldsymbol{V}_0\boldsymbol{X}^T + \boldsymbol{\Sigma} \end{pmatrix} \right).$$

Here $\boldsymbol{\Sigma}$ is the noise covariance, while $\boldsymbol{X}\boldsymbol{V}_0\boldsymbol{X}^T + \boldsymbol{\Sigma}$ is the marginal covariance of the observations. Conditioning this joint distribution on $\boldsymbol{Y} = \boldsymbol{y}$ gives the ordinary Gaussian posterior

$$\begin{aligned} \boldsymbol{a} \mid \boldsymbol{y} &\sim N(\boldsymbol{m}, \boldsymbol{V}), \\ \boldsymbol{m} &= \boldsymbol{m}_0 + \boldsymbol{V}_0\boldsymbol{X}^T(\boldsymbol{X}\boldsymbol{V}_0\boldsymbol{X}^T + \boldsymbol{\Sigma})^{-1}(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{m}_0), \\ \boldsymbol{V} &= \boldsymbol{V}_0 - \boldsymbol{V}_0\boldsymbol{X}^T(\boldsymbol{X}\boldsymbol{V}_0\boldsymbol{X}^T + \boldsymbol{\Sigma})^{-1}\boldsymbol{X}\boldsymbol{V}_0. \end{aligned} \tag{2}$$

These equations specify the Gaussian law that we will condition on shape, with factorizations and linear solves used in its numerical preparation.

We treat positivity, monotonicity, and convexity as sign restrictions of orders zero, one, and two. Write $C^0(I) = C(I)$ and $\frac{\mathrm{d}^0 f}{\mathrm{d}x^0} = f$. For $r \in \{0, 1, 2\}$ we condition the ordinary posterior on the function-space event

$$\mathcal{C}_r = \left\{ f \in C^r(I) : \min_{x \in I} \frac{\mathrm{d}^r f}{\mathrm{d}x^r}(x) \geq 0 \right\}. \tag{3}$$

The zero order gives a nonnegative response, the first gives nondecreasing regression, and the second gives convex regression, with $\mathcal{C}_r$ also denoting its explicitly identified preimage in coefficient space. The restriction requires nonnegativity but does not impose a fixed positive lower bound on the response, slope, or curvature.

Shape information can equivalently be incorporated into the prior. Provided that $\mathcal{C}_r$ has positive prior probability, we may restrict the Gaussian prior to curves in $\mathcal{C}_r$ and then update it using the observation likelihood. This gives the same posterior as first performing ordinary Gaussian updating and then conditioning on $\mathcal{C}_r$, because the prior normalizing constant cancels in Bayes' formula.

2.1 Continuum constraints through a Gaussian coefficient

To express a continuum restriction through one coefficient, we first choose a function in the retained span whose selected derivative, including the function itself at order zero, is positive throughout I. A sufficiently large coefficient along this function then makes the selected quantity nonnegative everywhere on the interval, whatever the values of the remaining coefficients. Since an invertible change of coordinates preserves every admissible coefficient vector in the finite space, the restriction can be represented by a single lower bound without changing the underlying Gaussian law. The following result integrates that coefficient and reconstructs it conditionally, using the standard normal distribution function Φ and survival function $\Phi^{\mathrm{c}}(x) = 1 - \Phi(x)$.

Theorem 1 (Gaussian conditioning under a continuum sign restriction)**.** *Under the finite Gaussian model in Section 2, let $\boldsymbol{a} = (a_0, \ldots, a_m)^T$ denote the coefficient vector and fix $r \in \{0, 1, 2\}$. Suppose an invertible linear change of coordinates $\boldsymbol{a} \mapsto (\boldsymbol{g}, \gamma) \in \mathbb{R}^m \times \mathbb{R}$ gives $f_{\boldsymbol{a}} = f_{\boldsymbol{g}} + \gamma h_r$. Here h_r is a fixed function in the retained span, chosen to have a uniformly positive rth derivative on I, and γ is its scalar coefficient. The curve $f_{\boldsymbol{g}}$ depends linearly on the remaining coordinates $\boldsymbol{g}$. Assume these functions belong to $C^r(I)$ with $\min_{x \in I} \frac{\mathrm{d}^r h_r}{\mathrm{d}x^r}(x) > 0$. In these coordinates, the ordinary Gaussian posterior has the form*

$$\boldsymbol{g} \mid \boldsymbol{y} \sim N(\boldsymbol{m}_g, \boldsymbol{V}_g), \qquad \gamma \mid \boldsymbol{g}, \boldsymbol{y} \sim N(\mu(\boldsymbol{g}), s^2). \tag{4}$$

Here $\boldsymbol{V}_g$ is positive definite, μ is affine, and $s > 0$ is constant in $\boldsymbol{g}$. The finite, attained threshold

$$\tau_r(\boldsymbol{g}) = \max_{x \in I} \frac{-\frac{\mathrm{d}^r f_{\boldsymbol{g}}}{\mathrm{d}x^r}(x)}{\frac{\mathrm{d}^r h_r}{\mathrm{d}x^r}(x)} \tag{5}$$

satisfies $f_{\boldsymbol{a}} \in \mathcal{C}_r$ exactly when $\gamma \geq \tau_r(\boldsymbol{g})$. This event has positive ordinary posterior probability and a null boundary in coefficient space. Its marginal density for $\boldsymbol{g}$ with respect to Lebesgue measure on $\mathbb{R}^m$ is

$$p(\boldsymbol{g} \mid \boldsymbol{y}, \mathcal{C}_r) = \frac{p(\boldsymbol{g} \mid \boldsymbol{y})\Phi^{\mathrm{c}}((\tau_r(\boldsymbol{g}) - \mu(\boldsymbol{g}))/s)}{P(f_{\boldsymbol{a}} \in \mathcal{C}_r \mid \boldsymbol{y})}. \tag{6}$$

Given this draw of $\boldsymbol{g}$, draw γ from the conditional normal in Equation (4) truncated below at $\tau_r(\boldsymbol{g})$ and apply the inverse coefficient map. Equivalently, for an independent $U \sim \mathrm{Uniform}(0,1)$,

$$\gamma = \mu(\boldsymbol{g}) - s\Phi^{-1}\big(U\Phi^{\mathrm{c}}((\tau_r(\boldsymbol{g}) - \mu(\boldsymbol{g}))/s)\big). \tag{7}$$

This reconstructs the full conditioned coefficient law, under which $\min_I \frac{\mathrm{d}^r f_{\boldsymbol{a}}}{\mathrm{d}x^r} > 0$ almost surely.

The Gaussian tail in Equation (6) is the conditional probability that the separated coefficient is large enough to satisfy the constraint, so values of $\boldsymbol{g}$ receive less posterior weight when they require an unlikely value of that coefficient. Conditional reconstruction retains the resulting dependence on $\boldsymbol{g}$ and therefore gives joint draws of the curve and its derivatives. This dependence is inherited from the original Gaussian model and must be transported through the coefficient map, since assigning a new independent prior to the transformed coordinates would change the law.

A related conditioning structure appears in GP regression with censored observations, where a Gaussian posterior density is weighted by the conditional probability of the censoring event (Gorm Hoffmann et al., 2025). In that setting, inequalities describe censored measurements at finitely many locations, whereas the threshold here represents a restriction on the latent curve throughout an interval.

The null-boundary statement in Theorem 1 means that, under its assumptions, the conditioned law assigns probability zero to curves whose minimum response or selected derivative equals zero. Thus a draw from this law has a strictly positive minimum almost surely, although these minima can be arbitrarily close to zero. For monotonicity, positivity of the derivative at every point is stronger than strict ordering of function values, which alone does not require a positive derivative everywhere. Nearly flat regions therefore remain possible, but the analyticity of the finite expansion means that assigning positive probability to exact plateaus or linear segments of positive length requires a different prior.

The random intercept supplies a constant direction for positivity, whereas monotonicity and convexity require directions with positive slope and curvature. For each choice, evaluating the threshold still requires finding a global extremum over I. On the padded interval of length $2L$, the zero boundary conditions require $\sin(2L\omega_j) = 0$, giving $\omega_j = j\pi/(2L) = j\omega_1$ for $j \geq 1$. The basis and its derivatives therefore involve sine and cosine harmonics of a common fundamental frequency. We introduce

$$\theta(x) = \omega_1(x - c + L), \qquad w(x) = \cos\theta(x),$$

because each cosine harmonic becomes a polynomial in w, while each sine harmonic becomes $\sin\theta$ times a polynomial in w, through identities for the Chebyshev polynomials in Definition 1. These identities allow the stationary locations needed for the thresholds to be characterized by polynomial equations.

Since I lies strictly inside the padded domain, $0 < \theta(x) < \pi$ and $\mathrm{d}w/\mathrm{d}x = -\omega_1 \sin\theta(x) < 0$ throughout I. Thus w maps I one-to-one onto a compact interval in $(-1, 1)$, reversing the order of its endpoints. Its nonzero derivative ensures that interior stationary locations correspond in the two coordinates, so the change of variable preserves the complete search for a continuum extremum.

Definition 1 (Chebyshev polynomials)**.** The Chebyshev polynomials of the first kind, T_j, and second kind, U_j, are defined for $w \in \mathbb{R}$ by the initial values and recurrences

$$\begin{aligned} T_0(w) &= 1, \quad T_1(w) = w, \quad T_{j+1}(w) = 2wT_j(w) - T_{j-1}(w), \\ U_0(w) &= 1, \quad U_1(w) = 2w, \quad U_{j+1}(w) = 2wU_j(w) - U_{j-1}(w), \end{aligned}$$

where the recurrences hold for integers $j \geq 1$.

For $w = \cos\theta$, these polynomials satisfy the standard trigonometric identities (DLMF, Equations (18.5.1) and (18.5.2))

$$T_j(w) = \cos(j\theta), \qquad \sin\theta\, U_{j-1}(w) = \sin(j\theta), \qquad j \geq 1.$$

In particular, the basis and its first two derivatives satisfy

$$\phi_{j,L}(x) = \frac{\sin\theta(x)}{\sqrt{L}}\, U_{j-1}(w(x)), \quad \frac{\mathrm{d}\phi_{j,L}}{\mathrm{d}x}(x) = \frac{j\omega_1}{\sqrt{L}}\, T_j(w(x)), \quad \frac{\mathrm{d}^2\phi_{j,L}}{\mathrm{d}x^2}(x) = -\omega_j^2\phi_{j,L}(x).$$

The second derivative therefore shares the sine factor in the first identity. For positivity, the polynomial representing the first derivative identifies the stationary locations of the curve. With the directions chosen below, monotonicity gives a ratio of polynomials with a positive denominator, while convexity gives a polynomial after the common sine factor cancels. Differentiating these expressions reduces their stationary-point conditions to polynomial equations. Evaluating the interval endpoints and these locations determines the global thresholds, with constant cases handled explicitly in the corollaries.

The following three corollaries apply Theorem 1 to positivity, monotonicity, and convexity, specifying a suitable direction and the corresponding continuum threshold for each restriction.

Corollary 1 (Positivity in the Laplacian expansion)**.** *Under the finite Gaussian model in Section 2, suppose $m \geq 1$ and $I \subset (c - L, c + L)$. Set $h_0 = 1$, $\boldsymbol{g} = (a_1, \ldots, a_m)^T$, and $f_{\boldsymbol{g}} = \sum_{j=1}^m a_j\phi_{j,L}$. Then $f_{\boldsymbol{a}} = f_{\boldsymbol{g}} + a_0h_0$, with inverse obtained by reinserting a_0.*

For $\theta(x) = \omega_1(x - c + L)$, $w(x) = \cos\theta(x)$, and $J = [w(x_+), w(x_-)] \subset (-1, 1)$, define $P_{0,\boldsymbol{g}}(w) = \sum_{j=1}^m ja_jT_j(w)$. The threshold is

$$\tau_0(\boldsymbol{g}) = -\min_{x\in I} f_{\boldsymbol{g}}(x). \tag{8}$$

Theorem 1 applies with scalar coefficient $\gamma = a_0$.

The threshold is attained at a physical endpoint or a location whose cosine coordinate is an interior real root in J of $P_{0,\boldsymbol{g}}$. This stationary polynomial has degree at most m when nonzero. All root multiplicities and reduced degrees are included. If the stationary polynomial vanishes identically, the curve $f_{\boldsymbol{g}}$ is constant and either endpoint suffices.

Corollary 2 (Monotonicity in the Laplacian expansion)**.** *Under the finite Gaussian model in Section 2, suppose $m \geq 4$ and $I \subset [c - L/2, c + L/2]$. Set $h_1 = -\phi_{2,L} - \phi_{4,L}/4$, $\boldsymbol{g} = (g_0, g_1, g_3, \ldots, g_m)^T$, and $f_{\boldsymbol{g}} = g_0 + \sum_{j=1}^m g_j\phi_{j,L}$, where $g_2 = 0$, $g_4 = a_4 - a_2/4$, and $g_j = a_j$ for $j \notin \{2, 4\}$. Then $f_{\boldsymbol{a}} = f_{\boldsymbol{g}} + (-a_2)h_1$, with inverse obtained by reinserting a_2 and setting $a_4 = g_4 + a_2/4$.*

For $\theta(x) = \omega_1(x - c + L)$, $w(x) = \cos\theta(x)$, and $J = [w(x_+), w(x_-)] \subset [-1/\sqrt{2}, 1/\sqrt{2}]$, define

$$P_{\boldsymbol{g}}(w) = \sum_{\substack{1\leq j\leq m \\ j\neq 2}} jg_jT_j(w), \qquad D(w) = 1 + 4w^2 - 8w^4. \tag{9}$$

Since $D \geq 1$ on J, the threshold is

$$\tau_1(\boldsymbol{g}) = \max_{w \in J} \frac{-P_{\boldsymbol{g}}(w)}{D(w)}. \tag{10}$$

Theorem 1 applies with scalar coefficient $\gamma = -a_2$.

The threshold is attained at a physical endpoint or a location whose cosine coordinate is an interior real root in J of $\frac{\mathrm{d}P_{\boldsymbol{g}}}{\mathrm{d}w} D - P_{\boldsymbol{g}} \frac{\mathrm{d}D}{\mathrm{d}w}$. This stationary polynomial has degree at most $m+3$ when nonzero. All root multiplicities and reduced degrees are included. If the stationary polynomial vanishes identically, the quotient $-P_{\boldsymbol{g}}/D$ is constant and either endpoint suffices. A nonzero constant stationary polynomial gives only endpoint candidates.

Corollary 3 (Convexity in the Laplacian expansion)**.** *Under the finite Gaussian model in Section 2, suppose $m \geq 2$ and $I \subset (c-L, c+L)$. Set $h_2 = -\phi_{1,L}$, $\boldsymbol{g} = (a_0, a_2, \ldots, a_m)^T$, and $f_{\boldsymbol{g}} = a_0 + \sum_{j=2}^{m} a_j \phi_{j,L}$. Then $f_{\boldsymbol{a}} = f_{\boldsymbol{g}} + (-a_1) h_2$, with inverse obtained by reinserting a_1.*

For $\theta(x) = \omega_1(x - c + L)$, $w(x) = \cos\theta(x)$, and $J = [w(x_+), w(x_-)] \subset (-1, 1)$, define the polynomial and threshold

$$Q_{\boldsymbol{g}}(w) = \sum_{j=2}^{m} j^2 a_j U_{j-1}(w), \qquad \tau_2(\boldsymbol{g}) = \max_{w \in J} Q_{\boldsymbol{g}}(w). \tag{11}$$

Theorem 1 applies with scalar coefficient $\gamma = -a_1$.

The threshold is attained at a physical endpoint or a location whose cosine coordinate is an interior real root in J of $\frac{\mathrm{d}Q_{\boldsymbol{g}}}{\mathrm{d}w}$. The polynomial $Q_{\boldsymbol{g}}$ has degree at most $m-1$, so its stationary polynomial has degree at most $m-2$ when nonzero. All root multiplicities and reduced degrees are included. If the stationary polynomial vanishes identically, $Q_{\boldsymbol{g}}$ is constant and either endpoint suffices. A nonzero constant stationary polynomial gives only endpoint candidates.

For positivity, separating a_0 leaves the spectral curve $f_{\boldsymbol{g}}$, and Equation (8) is the vertical shift needed to make its minimum nonnegative. The intercept's conditional posterior mean can depend on the spectral coefficients, so reconstruction uses that conditional law even though the prior coordinates are independent. The monotonicity map makes the same need to preserve dependence explicit by replacing a_4 with $g_4 = a_4 - a_2/4$, which is correlated with the separated coefficient $-a_2$. The convexity map simply sets aside the first mode, but all three constructions transport the ordinary Gaussian posterior before conditioning. In the two derivative cases, the intercept stays among the remaining coefficients and can change its posterior through correlation, despite contributing neither slope nor curvature.

Although all three constructions use the same padded basis domain $(c-L, c+L)$, their directions give different sufficient conditions on the physical interval. The constant positivity direction requires no additional padding, and h_2 has positive curvature throughout the open basis domain, whereas the slope of the chosen h_1 is bounded away from zero on the central half, including its endpoints. The conditions in Table 1 reflect these properties of the selected directions, so imposing the central-half condition on convexity would unnecessarily restrict that finite result.

To compute the thresholds in Equations (8), (10) and (11), we compare values at the two interval endpoints and at the interior stationary locations identified by the polynomial roots in J. For positivity, we evaluate $f_{\boldsymbol{g}}$ at the corresponding locations in I and take the negative of the smallest value. For monotonicity and convexity, we take the largest value of $-P_{\boldsymbol{g}}/D$ and $Q_{\boldsymbol{g}}$, respectively, at the candidate values of w. Every interior real root must be included, even when the derivative

Quantity	Positivity	Monotonicity	Convexity
Selected order	$r = 0$	$r = 1$	$r = 2$
Separated coefficient	a_0	$-a_2$	$-a_1$
Direction	1	$-\phi_{2,L} - \phi_{4,L}/4$	$-\phi_{1,L}$
Finite interval	$I \subset (c - L, c + L)$	$I \subset [c - L/2, c + L/2]$	$I \subset (c - L, c + L)$
Rank condition	$m \geq 1$	$m \geq 4$	$m \geq 2$
Threshold	$-\min_I f_{\boldsymbol{g}}$	$\max_J(-P_{\boldsymbol{g}}/D)$	$\max_J Q_{\boldsymbol{g}}$
Polynomial degree bound	m	$m + 3$	$m - 2$
Limit space	$C(I)$	$C^1(I)$	$C^2(I)$
Matérn smoothness	$\nu > 0$	$\nu > 1$	$\nu > 2$

Table 1: Positivity, monotonicity, and convexity as alternative conditioning events in the same spectral model. Here I is the physical interval and J its image under the decreasing cosine coordinate w. The degree bounds apply to the nonzero polynomials whose roots identify stationary locations. The finite rank and interval conditions are sufficient for the displayed directions. The function-space limits additionally require Assumption 2 and the padding and frequency-cutoff limits in Theorem 2.

does not change sign there. If the stationary polynomial vanishes identically, the function being optimized is constant and an endpoint evaluation suffices.

Nonpositivity, nonincreasing regression, and concavity follow by negating the observations and prior mean while retaining all covariances, then applying the corresponding nonnegative-derivative construction and negating the reconstructed curve and its derivatives. In particular, concavity makes the slope nonincreasing but allows it to have either sign, so a curvature restriction alone does not impose monotonicity. Table 2 summarizes the inverse maps for all six individual sign constraints.

Constraint	Reinserted coefficients	Remaining coefficients
Positivity	$a_0 = \gamma$	$a_j = g_j$, $1 \leq j \leq m$
Negativity	$a_0 = -\gamma$	$a_j = -g_j$, $1 \leq j \leq m$
Nondecreasing	$a_2 = -\gamma$, $a_4 = g_4 - \gamma/4$	$a_j = g_j$, $j \notin \{2, 4\}$
Nonincreasing	$a_2 = \gamma$, $a_4 = -g_4 + \gamma/4$	$a_j = -g_j$, $j \notin \{2, 4\}$
Convexity	$a_1 = -\gamma$	$a_j = g_j$, $j \neq 1$
Concavity	$a_1 = \gamma$	$a_j = -g_j$, $j \neq 1$

Table 2: Inverse maps to the original coefficients for six constraints on I. Positivity and negativity mean $f_{\boldsymbol{a}} \geq 0$ and $f_{\boldsymbol{a}} \leq 0$. The negative, nonincreasing, and concave cases use $(\boldsymbol{g}, \gamma)$ from the Gaussian calculation for $-f_{\boldsymbol{a}}$ with negated observations and prior mean. The other cases use $f_{\boldsymbol{a}}$. Indices range from 0 to m unless stated otherwise, and g_j retains its original coefficient index. Each pair uses the rank and padding conditions of its corresponding case in Corollaries 1 to 3.

Several restrictions can describe different aspects of the same response. A nonnegative, nondecreasing, concave curve, for example, has a nonnegative level and slope, with the slope nonincreasing across the interval. We express such intersections by selecting derivative orders and a sign at each order. For a nonempty index set $A \subseteq \{0, 1, 2\}$ and a sign choice $\varepsilon : A \to \{-1, 1\}$, write $\varepsilon_r = \varepsilon(r)$ and define

$$\mathcal{C}_A^{\varepsilon} = \left\{ f \in C^{\max A}(I) : \varepsilon_r \frac{\mathrm{d}^r f}{\mathrm{d}x^r}(x) \geq 0 \text{ for every } r \in A \text{ and } x \in I \right\}. \tag{12}$$

At order zero, the two signs give nonnegativity and nonpositivity. At orders one and two, they give nondecreasingness and nonincreasingness, and convexity and concavity, respectively. Taking $\varepsilon_r = 1$ at every selected order gives positivity, nondecreasingness, and convexity, their three pairwise combinations, and their triple combination. Allowing either sign gives six individual cases and 20 joint combinations without opposing signs. Imposing both signs at the same order instead forces the corresponding derivative to vanish throughout I, an event of zero probability under the finite Gaussian model.

To retain the scalar conditioning construction, we need one direction h in the retained span whose selected signed derivatives are all uniformly positive on I. Increasing its coefficient then increases every signed derivative involved in the restriction. Each order and location imposes a lower bound on this same coefficient, so their largest bound enforces the entire intersection. The direction condition must be verified for the chosen combination, rank, and interval. The following proposition gives the resulting Gaussian conditioning law.

Proposition 1 (Joint sign constraints)**.** *Under the finite Gaussian model in Section 2, let* $\boldsymbol{a} = (a_0, \ldots, a_m)^T$ *denote the coefficient vector and fix a nonempty* $A \subseteq \{0, 1, 2\}$ *and signs* $\varepsilon : A \to \{-1, 1\}$*, with event* $\mathcal{C}_A^\varepsilon$ *defined in Equation* (12)*. Suppose an invertible linear change of coordinates* $\boldsymbol{a} \mapsto (\boldsymbol{g}, \gamma) \in \mathbb{R}^m \times \mathbb{R}$ *gives* $f_{\boldsymbol{a}} = f_{\boldsymbol{g}} + \gamma h$*. Here* h *is a fixed function in the retained span, shared by all selected orders,* γ *is its scalar coefficient, and* $f_{\boldsymbol{g}}$ *depends linearly on the remaining coordinates* $\boldsymbol{g}$*. Assume* $h, f_{\boldsymbol{g}} \in C^{\max A}(I)$ *with* $\min_{x \in I} \varepsilon_r \frac{\mathrm{d}^r h}{\mathrm{d}x^r}(x) > 0$ *for every* $r \in A$*. Define*

$$\eta_r(x; \boldsymbol{g}) = \frac{-\varepsilon_r \frac{\mathrm{d}^r f_{\boldsymbol{g}}}{\mathrm{d}x^r}(x)}{\varepsilon_r \frac{\mathrm{d}^r h}{\mathrm{d}x^r}(x)}, \qquad \tau_A^\varepsilon(\boldsymbol{g}) = \max_{r \in A} \max_{x \in I} \eta_r(x; \boldsymbol{g}). \tag{13}$$

This finite, attained threshold satisfies $f_{\boldsymbol{a}} \in \mathcal{C}_A^\varepsilon$ *exactly when* $\gamma \geq \tau_A^\varepsilon(\boldsymbol{g})$*. The event has positive ordinary posterior probability and a null coefficient boundary. Its marginal density and reconstruction are Equations* (6) *and* (7)*, with* $\tau_r, \mathcal{C}_r$ *replaced by* $\tau_A^\varepsilon, \mathcal{C}_A^\varepsilon$ *and the Gaussian parameters in Equation* (4) *computed under this map. The conditioned law satisfies* $\min_{x \in I} \varepsilon_r \frac{\mathrm{d}^r f_{\boldsymbol{a}}}{\mathrm{d}x^r}(x) > 0$ *almost surely for every* $r \in A$*. The envelope, strong log-concavity, and conservative threshold-error conclusions extend under their respective hypotheses.*

The joint restrictions can also reduce the number of continuum searches. For the nonnegative, nondecreasing, concave example, take $A = \{0, 1, 2\}$ with $\varepsilon_0 = \varepsilon_1 = 1$ and $\varepsilon_2 = -1$. Concavity makes the slope smallest at x_+, and nondecreasingness makes the response smallest at x_-. Under the common map in Proposition 1, the threshold therefore becomes

$$\tau_A^\varepsilon(\boldsymbol{g}) = \max \left\{ \max_{x \in I} \eta_2(x; \boldsymbol{g}), \eta_1(x_+; \boldsymbol{g}), \eta_0(x_-; \boldsymbol{g}) \right\}.$$

Thus this combination requires one continuum curvature bound and two endpoint bounds. All three use the same h and $\boldsymbol{g}$, whose choice determines the quotients in Equation (13). The supplementary material gives the HSGP coefficient map, conditions on its direction coefficients, and the reductions for all sign choices, with a table separating continuum and endpoint bounds for the 20 joint cases.

3 Geometry, numerical error, and approximation

At a fixed location $x \in I$, the ratio in Equation (5) is the lower bound on γ required to satisfy the sign restriction at that location. Because $f_{\boldsymbol{g}}$ depends linearly on $\boldsymbol{g}$ and h_r is fixed, this bound is a linear function $\boldsymbol{\beta}_r(x)^T \boldsymbol{g}$. Here $\boldsymbol{\beta}_r(x) \in \mathbb{R}^m$ is the vector of coefficient multipliers, with its

entries ordered as those of $\boldsymbol{g}$. For positivity, for example, $h_0 = 1$ and $\boldsymbol{g} = (a_1, \ldots, a_m)^T$, so $\boldsymbol{\beta}_0(x) = (-\phi_{1,L}(x), \ldots, -\phi_{m,L}(x))^T$. The continuum threshold is therefore

$$\tau_r(\boldsymbol{g}) = \max_{x \in I} \boldsymbol{\beta}_r(x)^T \boldsymbol{g}.$$

The continuity assumptions and positive derivative bound in Theorem 1 make $\boldsymbol{\beta}_r$ continuous and bounded on I, so the maximum representation makes τ_r convex, positively homogeneous, and globally Lipschitz. The threshold enters the log posterior through the normal survival weight, so its derivatives are needed to compute the gradients used in posterior sampling. Danskin's theorem (Danskin, 1966) expresses these derivatives through the coefficient vectors at maximizing locations. When the maximizing location $x_\star$ is unique, the threshold has gradient $\boldsymbol{\beta}_r(x_\star)$. Several maximizing locations also permit a gradient when their coefficient vectors agree, whereas different vectors produce kinks. The supplementary material gives the threshold derivatives and the resulting log-posterior gradient formulas. For a joint restriction in Proposition 1, the vectors are formed under the common coefficient map and the maximum ranges over pairs $(r, x) \in A \times I$. A change in the maximizing derivative order can therefore produce a kink as well as a change in its location.

Let $\boldsymbol{B}\boldsymbol{B}^T = \boldsymbol{V}_g$ with $\boldsymbol{B}$ invertible, and write $\boldsymbol{g}(\boldsymbol{z}) = \boldsymbol{m}_g + \boldsymbol{B}\boldsymbol{z}$. The exact whitened log density, up to a constant, is

$$-\tfrac{1}{2}\|\boldsymbol{z}\|^2 + \log \Phi^{\mathrm{c}}\left(\frac{\tau_r(\boldsymbol{g}(\boldsymbol{z})) - \mu(\boldsymbol{g}(\boldsymbol{z}))}{s}\right). \tag{14}$$

The supplementary material establishes that this density is 1-strongly log-concave, giving a unique mode and finite moments of every polynomial order.

The continuum maximum generally requires numerical evaluation, and an underestimated threshold would admit curves that violate the intended sign restriction. We therefore use a conservative upper threshold $\widetilde{\tau}_r \geq \tau_r$, so that imposing $\gamma \geq \widetilde{\tau}_r(\boldsymbol{g})$ preserves the restriction. The cost is that admissible coefficient values between the exact and upper thresholds are excluded from the posterior. The upper threshold may not be convex, so the log-concavity conclusion need not carry over to this numerical posterior. Its difference from the exact law can instead be quantified through the probability mass of the excluded coefficients, as the following proposition shows under exact Gaussian operations.

Proposition 2 (Posterior error from a conservative continuum threshold)**.** *Under Theorem 1, let $\widetilde{\tau}_r$ be measurable and satisfy $\tau_r \leq \widetilde{\tau}_r \leq \tau_r + \delta$ almost everywhere under the ordinary posterior of $\boldsymbol{g}$, for a fixed $\delta \geq 0$. Use the same upper threshold in marginal weighting and reconstruction. Let $P_{\boldsymbol{y}}$ be the ordinary coefficient posterior, and use $\mathcal{C}_r$ also for its coefficient preimage. Define $\widetilde{\mathcal{C}}_r = \{\gamma \geq \widetilde{\tau}_r(\boldsymbol{g})\}$. Both conditioned laws exist and, with* $\mathrm{TV}(P, Q) = \sup_A |P(A) - Q(A)|$,

$$\mathrm{TV}(P_{\boldsymbol{y}}(\cdot \mid \widetilde{\mathcal{C}}_r), P_{\boldsymbol{y}}(\cdot \mid \mathcal{C}_r)) = 1 - \frac{P_{\boldsymbol{y}}(\widetilde{\mathcal{C}}_r)}{P_{\boldsymbol{y}}(\mathcal{C}_r)} \leq \min\left(1, \frac{\delta}{s\sqrt{2\pi} P_{\boldsymbol{y}}(\mathcal{C}_r)}\right). \tag{15}$$

Every measurable summary of the reconstructed coefficients inherits this bound. The identity holds for any inner event of positive probability.

The conditioning probability in Equation (15) explains why a rare shape event is more sensitive to a given absolute threshold error. The bound requires the threshold guarantee to hold globally or almost everywhere under the ordinary posterior and quantifies the effect of threshold arithmetic within the finite model, leaving basis approximation and finite-chain error as separate quantities.

3.1 Convergence to the conditioned underlying GP

We next consider the error introduced by replacing the underlying GP with a finite HSGP expansion, and how this error depends on padding and rank. Increasing L separates the artificial boundary from the physical interval I, whereas m/L controls the frequency cutoff, so increasing padding at fixed rank can remove variation at short scales. The two choices must therefore be considered together and for the derivative used in inference as well as for the response (Riutort-Mayol et al., 2023).

Because the conditioning event depends on the response or its selected derivative between grid points, convergence of covariance blocks on a finite grid is insufficient for comparing the conditioned laws. We instead work on the entire physical interval in $C^r(I)$ with its Borel sigma-field and norm

$$\|f\|_{C^r} = \sum_{j=0}^{r} \sup_{x\in I} \left| \frac{\mathrm{d}^j f}{\mathrm{d}x^j}(x) \right|, \qquad \frac{\mathrm{d}^0 f}{\mathrm{d}x^0} = f.$$

Assumption 1 (Common underlying and finite Gaussian models)**.** Fix $I \subset [c-R, c+R]$ with $R > 0$ and the observation model and intercept law specified in Section 2. Let $F = a_0 + W$, where W is a centered stationary GP on $\mathbb{R}$, independent of a_0, with covariance function k and spectral density S_k as defined in Section 2. Use that same spectrum and intercept in the finite models in Equation (1), varying only m and L while keeping the center, interval, spectrum, intercept law, observation locations, positive-definite noise covariance, and data identical in all models.

Assumption 2 (Spectral regularity for the selected order)**.** Fix $r \in \{0, 1, 2\}$. The spectrum S_k is continuous, even, and strictly positive. There is a regularity parameter $\zeta \in (0, 1]$ such that the two functions

$$\omega \longmapsto S_k(\omega), \qquad \omega \longmapsto \omega^{2r+2\zeta} S_k(\omega)$$

are locally absolutely continuous on $[0, \infty)$, and each function and its derivative belongs to $L^1(0, \infty)$. The weighted spectrum $\omega^{2r+2\zeta} S_k(\omega)$ tends to zero as $\omega \downarrow 0$, and we use this limiting value at the origin. The frequency exponent is 2ζ for positivity, $2 + 2\zeta$ for monotonicity, and $4 + 2\zeta$ for convexity.

Under the Fourier representation of the stationary covariance, differentiating r times with respect to each input multiplies the spectrum by ω^{2r}. The additional factor $\omega^{2\zeta}$ allows us to bound the variance of the difference between the rth derivative values at x and x' by a constant multiple of $|x - x'|^{2\zeta}$. With ζ held fixed in the approximation limit, this increment bound supplies the additional regularity needed for continuity and tightness in $C^r(I)$.

Let $\Pi^{\mathcal{C}_r}_{m,L,\boldsymbol{y}}$ and $\Pi^{\mathcal{C}_r}_{\boldsymbol{y}}$ denote the finite and underlying posterior laws obtained by weighting their priors by the common Gaussian likelihood and the indicator of Equation (3). For laws with finite second norm moments, define

$$W_2(P, Q) = \inf_{\lambda} \left(\int \|f - g\|^2_{C^r} \, \lambda(df, dg) \right)^{1/2},$$

where λ ranges over couplings with marginals P, Q.

Convergence in W_2 means that the finite-model curve $F_{m,L}$ and the underlying curve F, with their respective posterior distributions conditional on the data and $\mathcal{C}_r$, can be coupled so that

$$\mathrm{E}\left[\sup_{x\in I} \left| \frac{\mathrm{d}^j F_{m,L}}{\mathrm{d}x^j}(x) - \frac{\mathrm{d}^j F}{\mathrm{d}x^j}(x) \right|^2 \right] \longrightarrow 0, \qquad j = 0, \ldots, r,$$

as $L \to \infty$ and $m/L \to \infty$. Under this coupling, the expected squared maximum discrepancy over I vanishes for the response and each derivative through order r.

Theorem 2 (Convergence of shape-constrained HSGP posteriors)**.** *Fix $r \in \{0, 1, 2\}$ under Assumptions 1 and 2, and consider finite models with $L \geq 2R$ and $m \geq 1$ for $r = 0$, $m \geq 4$ for $r = 1$, or $m \geq 2$ for $r = 2$. The underlying GP has a $C^r(I)$ version, and the finite and underlying shape events have positive probability and null boundaries in that space. Consequently, the posteriors conditional on the data and $\mathcal{C}_r$ exist, with finite moments of every polynomial order in $\|\cdot\|_{C^r}$. As $L \to \infty$ and $m/L \to \infty$,*

$$W_2(\Pi^{\mathcal{C}_r}_{m,L,\boldsymbol{y}}, \Pi^{\mathcal{C}_r}_{\boldsymbol{y}}) \longrightarrow 0. \tag{16}$$

This convergence also holds for the posterior summaries: means of $\frac{\mathrm{d}^j f}{\mathrm{d}x^j}$ converge uniformly on I for every $0 \leq j \leq r$, and covariances between $\frac{\mathrm{d}^{r_1} f}{\mathrm{d}x^{r_1}}(x)$ and $\frac{\mathrm{d}^{r_2} f}{\mathrm{d}x^{r_2}}(x')$ converge uniformly on $I \times I$ for every $0 \leq r_1, r_2 \leq r$. Under the conditioned underlying law, $\min_I \frac{\mathrm{d}^r f}{\mathrm{d}x^r} > 0$ almost surely.

The spectral weight controls the highest derivative in the function-space norm, so positivity needs only a continuous process, whereas monotonicity and convexity require C^1 and C^2 regularity, respectively. The proof adapts the derivative-covariance argument of Jensen and Hoffmann (2026, Theorem 2 and its proof) to the weighted-spectral conditions above, then uses increment moments to obtain convergence in that space and passes through likelihood and event conditioning. This last step requires positive event probability and a null boundary, which the independent random intercept supplies for positivity and a positive reproducing kernel Hilbert space (RKHS) derivative direction supplies for the two derivative constraints.

Squared exponential spectra satisfy all three sets of conditions. A Matérn spectrum is proportional to $(\lambda^2 + \omega^2)^{-(\nu+1/2)}$ with $\lambda > 0$ (Rasmussen and Williams, 2006, Chapter 4). Its weight of order d is integrable when $d < 2\nu$, and the weighted derivative is integrable for the orders used here, so any $0 < \zeta < \min(1, \nu - r)$ verifies Assumption 2 when $\nu > r$. The conditions also hold for positive finite mixtures with a common smaller exponent. A squared exponential covariance multiplied by $\cos(\omega_0 u)$, with fixed $\omega_0 \geq 0$, also qualifies because its spectrum is a positive sum of translated Gaussian densities.

4 Posterior computation

Integrating out γ gives a posterior density for the remaining coefficients $\boldsymbol{g}$. We sample from this density and then draw γ from its conditional truncated normal distribution to recover the full coefficient vector. We first form the ordinary Gaussian posterior in Equation (2) and apply the appropriate invertible map from Corollaries 1 to 3, or the common map in Proposition 1, to its mean and covariance, giving the Gaussian laws

$$\boldsymbol{g} \mid \boldsymbol{y} \sim N(\boldsymbol{m}_g, \boldsymbol{V}_g), \qquad \gamma \mid \boldsymbol{g}, \boldsymbol{y} \sim N(\mu(\boldsymbol{g}), s^2),$$

where μ is affine and $s > 0$ is constant in $\boldsymbol{g}$. These Gaussian parameters can therefore be prepared before marginal sampling, together with an invertible factor $\boldsymbol{B}\boldsymbol{B}^T = \boldsymbol{V}_g$ defining the whitened coordinates $\boldsymbol{g} = \boldsymbol{m}_g + \boldsymbol{B}\boldsymbol{z}$, in which $\boldsymbol{z}$ has the standard normal law on $\mathbb{R}^m$ under the ordinary posterior. Mapping through $\boldsymbol{B}$ preserves the Gaussian dependence among the remaining coefficients.

At each proposed $\boldsymbol{z}$, evaluating the marginal density requires forming $\boldsymbol{g}$ and its continuum threshold $\tau_r(\boldsymbol{g})$. For the individual cases, this means comparing both endpoints and all interior stationary candidates from Corollaries 1 to 3, or obtaining bounds over the entire interval. The Gaussian log-survival term in Equation (14) then weights the standard normal density by the conditional probability that γ satisfies the sign restriction. Since the observation likelihood is already included in the ordinary posterior and the overall conditioning probability is constant in $\boldsymbol{z}$,

neither requires a separate evaluation during marginal sampling. The monotonicity and convexity implementations sample this unrestricted m-dimensional density by Hamiltonian Monte Carlo, with the numerical threshold evaluation described below.

For each retained $\boldsymbol{g}$, we recover the separated coefficient by drawing γ from $N(\mu(\boldsymbol{g}), s^2)$ truncated to $[\tau_r(\boldsymbol{g}), \infty)$, using an independent $U \sim \text{Uniform}(0,1)$ and the inverse-survival formula in Equation (7), which retains its conditional uncertainty. Applying the corresponding inverse map in Table 2 then recovers the original coefficients of $f_{\boldsymbol{a}}$. With the exact threshold and draws from the exact marginal law, these two stages recover the full conditioned coefficient law, so evaluating the expansion and its derivatives gives joint posterior curves from which to compute means and pointwise credible intervals.

For a signed joint restriction satisfying Proposition 1, the common map determines both the Gaussian preparation and every quotient $\eta_r(x; \boldsymbol{g})$. At each proposed $\boldsymbol{g}$, we evaluate the required continuum and endpoint bounds and take their maximum to obtain $\tau_A^{\varepsilon}(\boldsymbol{g})$. Since these bounds all concern the same γ, a single survival probability $\Phi^{\mathrm{c}}((\tau_A^{\varepsilon}(\boldsymbol{g}) - \mu(\boldsymbol{g}))/s)$ weights the marginal density for the entire intersection. Reconstruction uses this joint lower bound in the truncated normal and then inverts the common coefficient map. The Gaussian parameters must be computed under that map, and the individual thresholds in Table 1, which use different maps, cannot be combined directly. The supplementary material specifies the HSGP map and the continuum and endpoint bounds for each joint combination.

In numerical evaluation, a successful threshold calculation returns an enclosure whose upper endpoint $\widetilde{\tau}_r$ satisfies

$$\tau_r(\boldsymbol{g}) \le \widetilde{\tau}_r(\boldsymbol{g}) \le \tau_r(\boldsymbol{g}) + \delta$$

for the supplied coefficients and a chosen tolerance $\delta > 0$. If marginal weighting and conditional reconstruction use the same numerical threshold and the enclosure bound holds almost everywhere under the ordinary posterior, Proposition 2 controls the total variation distance between the resulting target distribution and the posterior conditioned on the exact sign event. The conditional standard deviation s provides a natural scale for choosing δ, because δ/s bounds the error in the standardized truncation point. For a prescribed total variation error $\rho \in (0,1)$, Proposition 2 gives the sufficient choice

$$\delta \le \rho\, s\sqrt{2\pi}\, P_{\boldsymbol{y}}(\mathcal{C}_r).$$

In practice, a valid positive lower bound on the conditioning probability can replace $P_{\boldsymbol{y}}(\mathcal{C}_r)$ on the right-hand side. Without such a bound, a small value of δ/s controls the standardized threshold error but does not by itself certify a small posterior error, particularly when the shape event is rare.

Stable evaluation uses log-survival and inverse-survival calculations to avoid subtracting nearly equal probabilities in the tails. For gradient-based sampling, the derivative implementations evaluate approximate envelope gradients at numerically selected maximizing locations. At a unique maximum these gradients approximate derivatives of the exact threshold, while the sampled log values use the rounded upper threshold. The supplementary material explains how the numerical values and approximate gradients enter the Stan No-U-Turn sampler (NUTS), alongside the compact algorithm, Gaussian block formulas, threshold enclosures, precision controls, and stable reconstruction.

5 Applications

We compare ordinary Gaussian and shape-constrained posteriors in two applications where derivatives have direct scientific interpretations. Monotonicity ensures nonnegative mercury concentration in a lidar model, while concavity represents a nonincreasing marginal gain in leaf carbon assimilation.

5.1 Atmospheric mercury concentration

To illustrate how monotonicity changes derivative inference, we use all 221 released mercury lidar observations over 390–720 meters near the Bella Vista geothermal power station in Italy, distributed with the R package SemiPar (Holst et al., 1996; Feng et al., 2022). Each response is the log ratio of reflected on-resonant and off-resonant laser signals, with approximately 100 pulse pairs averaged during acquisition. These data have also been analyzed with a globally monotone GP approximation by Maatouk et al. (2025a).

The shape restriction follows from the differential absorption lidar (DIAL) approximation, which neglects differential backscatter and attenuation from other species. Under this approximation, the derivative of the mean log ratio per meter equals mercury concentration multiplied by -1.6×10^{-5} square meters per nanogram. Nonnegative concentration therefore requires a nondecreasing latent mean for the negative log ratio, even if the noisy observations are not ordered.

Write r for distance in meters, set $x(r) = (r - 390)/330$, and let Y_i be the negative recorded log ratio. We use 128 basis functions with center $c = 1/2$ and half-width $L = 2$, so the computational interval $[-1.5, 2.5]$ contains the physical interval $[0, 1]$. The Matérn 7/2 spectral prior in Equation (1) gives the hierarchy

$$
\begin{aligned}
f_{\boldsymbol{a}}(x) &= a_0 + \sum_{j=1}^{128} a_j \phi_{j,2}(x), \qquad a_0 \sim N(\mu_0, \sigma_0^2), \\
a_j &\overset{\text{ind}}{\sim} N\left(0, \sigma_f^2 \frac{32\lambda^7}{5(\lambda^2 + \omega_j^2)^4}\right), \qquad \lambda = \frac{\sqrt{7}}{\ell}, \\
Y_i \mid \boldsymbol{a} &\overset{\text{ind}}{\sim} N(f_{\boldsymbol{a}}(x(r_i)), \sigma_i^2).
\end{aligned}
$$

The intercept is independent of the other coefficients, with prior mean $\mu_0 = 0.291156$ and standard deviation $\sigma_0 = 1.952921$. These values are the observed response mean and twice its range, respectively. The underlying GP has marginal standard deviation $\sigma_f = 0.308590$ and normalized length scale $\ell = 0.288869$, corresponding to 95.33 meters.

Following the noise family of Holst et al. (1996), we allow measurement variance to change with distance by setting

$$
\sigma_i = \sigma_{\text{mid}} \exp\big(\gamma_1 (x(r_i) - c) + \gamma_2 (x(r_i) - c)^2\big).
$$

Here $\sigma_{\text{mid}} = 0.056125$ is the noise standard deviation at the midpoint of 555 meters, while $\gamma_1 = 2.330252$ and $\gamma_2 = -0.305387$ describe the linear and quadratic changes in its logarithm. All standard deviations are on the negative-logratio scale. We estimate the covariance and noise parameters jointly by ordinary underlying-GP marginal likelihood and use the same estimates in both posterior fits.

The inferential targets are the latent negative log ratio $f_{\boldsymbol{a}}(x(r))$ and the implied concentration

$$
\frac{1}{330(1.6 \times 10^{-5})} \left. \frac{\mathrm{d} f_{\boldsymbol{a}}}{\mathrm{d}x} \right|_{x = x(r)} \quad \text{nanograms per cubic meter.}
$$

The factor 330 converts the derivative in normalized coordinates to a derivative per meter, so conditioning on $\frac{\mathrm{d} f_{\boldsymbol{a}}}{\mathrm{d}x}(x) \geq 0$ throughout $[0, 1]$ imposes nonnegative concentration over the observed range. We use the procedure in Section 4 for the monotone posterior and evaluate the unconstrained Gaussian summaries analytically. Sampler settings, convergence diagnostics, and the full posterior predictive assessment are given in the supplementary material.

Figure 1 compares concentration inference with and without monotonicity under the same Gaussian model. At 502.86 meters, the ordinary posterior mean concentration reaches its minimum

on the reconstruction grid at -42.13 nanograms per cubic meter, with pointwise 95% equal-tail credible interval $(-94.27, 10.00)$ and posterior probability 0.9434 of a negative concentration. Conditioning on monotonicity gives a mean concentration of 11.75 nanograms per cubic meter at that distance, with corresponding interval $(1.10, 31.23)$, while changing the response fit little.

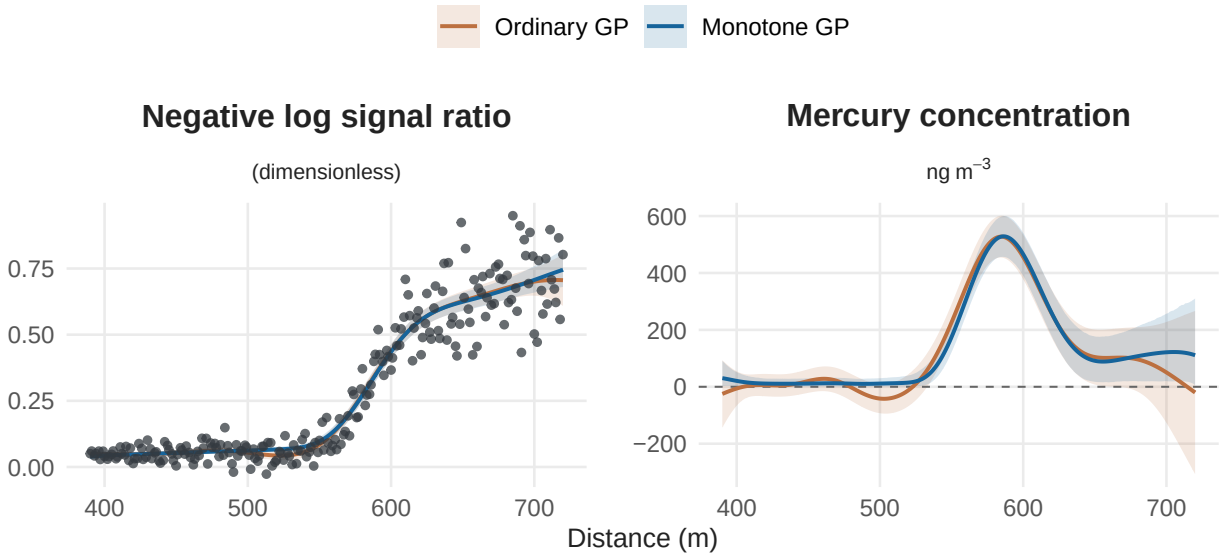


Figure 1: The left panel shows the latent negative log signal ratio with all 221 observations as points, and the right shows mercury concentration in nanograms per cubic meter, given under the DIAL approximation by the range derivative of the latent negative log ratio divided by 1.6×10^{-5} square meters per nanogram. Orange and blue lines denote ordinary and globally monotone posterior means, with shading for pointwise 95% equal-tail latent credible intervals excluding new measurement error. Both fits use the same rank-128 Matérn 7/2 Gaussian regression model and range-dependent noise variances, with the monotone posterior imposing the restriction throughout 390–720 meters.

5.2 Marginal carbon assimilation in a leaf light response

Concavity of a leaf light-response curve means that the marginal gain in carbon assimilation is nonincreasing with irradiance. We examine what this assumption contributes to inference from one response sequence in the Arctic gas-exchange study of Rogers et al. (2019b), using the measurements released by Rogers et al. (2019a). Each record is a chamber measurement of net assimilation on the same leaf at a specified incident irradiance. Before fitting either model, we select the smallest sample identifier in the release, which corresponds to an Eriophorum angustifolium leaf. The study's quality filters and exclusion of the dark measurement leave 13 records over irradiance 19.1–2,000.2 μmol quanta $\mathrm{m}^{-2}\mathrm{s}^{-1}$. The measurements were taken in descending illumination over approximately 37 minutes, with retained leaf temperatures between 14.1 and 15.0 °C.

Let $A(q)$ denote mean net assimilation at irradiance q under the chamber conditions, measured in μmol CO_2 $\mathrm{m}^{-2}\mathrm{s}^{-1}$. We compare the ordinary Gaussian posterior with its restriction to $\frac{\mathrm{d}^2 A}{\mathrm{d}q^2}(q) \leq 0$

over the observed irradiance range. In the scaled coordinate $x = (q - q_0)/d$, q_0 is the midpoint of that range and d its half-width, so the physical interval becomes $[-1, 1]$. The shared model is

$$\begin{aligned} A(q) &= a_0 + \sum_{j=1}^{24} a_j \phi_{j,2}((q - q_0)/d), \qquad a_0 \sim N(0, 20^2), \\ a_j &\overset{\text{ind}}{\sim} N\left(0, \sigma_f^2 \sqrt{2\pi}\ell \exp(-\ell^2 \omega_j^2/2)\right), \\ y_i \mid A &\overset{\text{ind}}{\sim} N(A(q_i), \sigma_\varepsilon^2). \end{aligned} \tag{17}$$

The intercept is independent of the other coefficients, and the basis center is zero in the scaled coordinate. An ordinary Gaussian marginal-likelihood calculation gives $\sigma_f = 5.625$, $\ell = 0.3034$, and $\sigma_\varepsilon = 0.3703$, which both posterior fits use to isolate the effect of the curvature restriction. The length scale corresponds to about 301 irradiance units, and the residual scale has assimilation units.

The inferential targets are the response $A(q)$, its marginal gain $\frac{\mathrm{d}A}{\mathrm{d}q}(q)$, and changes in that gain within the observed range. We report derivatives on the physical irradiance scale, applying factors $1/d$ and $1/d^2$ to first and second derivatives in the scaled coordinate. The concavity restriction makes marginal gain nonincreasing while allowing it to become negative, and imposes no finite saturation level beyond the fitted interval.

We fit the concave model by applying the convexity sampler to $-A$ and the negated responses, and obtain the unconstrained comparison by direct Gaussian simulation. Sampling settings, convergence diagnostics, and posterior predictive assessment are given in the supplementary material.

The response and derivative panels in Figure 2 show different effects of the constraint. Posterior mean responses are close at many observed irradiances, but the unconstrained marginal gain can increase within the range and has substantially wider pointwise intervals at high irradiance. Concavity imposes nonincreasing marginal gain on each posterior curve, so the ordering holds for the individual curves whose uncertainty the right panel summarizes.

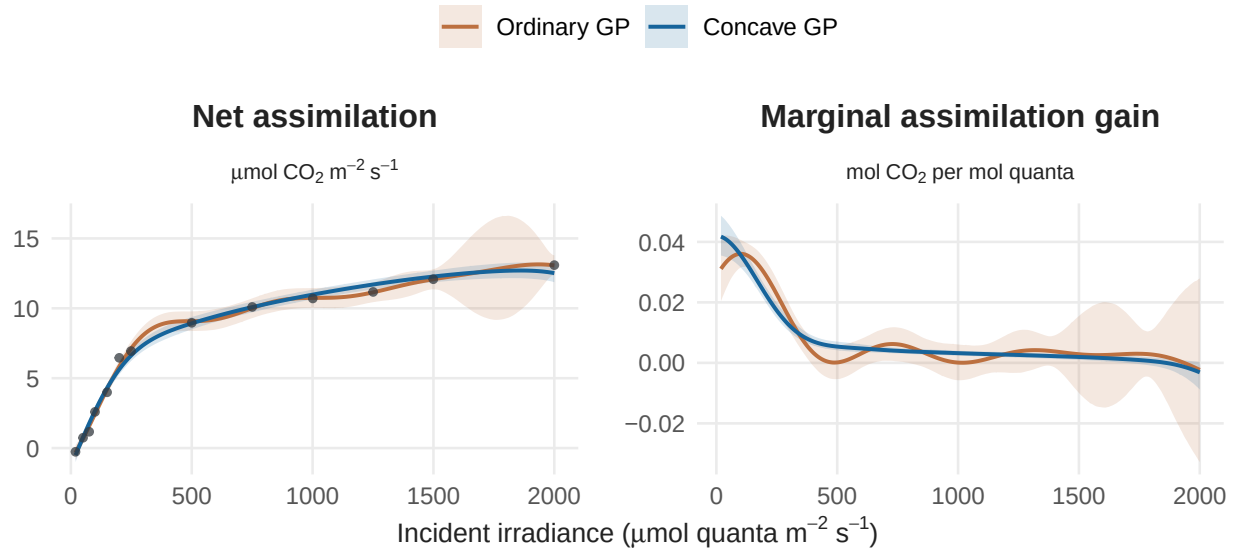


Figure 2: Posterior comparison for 13 net assimilation measurements from one Arctic Eriophorum angustifolium leaf. The left panel shows net assimilation in μmol CO_2 $m^{-2}s^{-1}$, with observations in black. The right panel shows its derivative with respect to incident irradiance, in mol CO_2 per mol quanta. Orange and blue lines are ordinary Gaussian and concavity-conditioned posterior means, with shaded pointwise 95% equal-tail credible intervals for the latent quantities. Both fits use the same covariance and noise parameters and the same rank-24 HSGP expansion.

Table 3 gives selected response and derivative summaries. For example, the assimilation increase from 500 to 1,500 irradiance units has concave posterior mean 3.35 and 95% equal-tail interval $(2.73, 3.99)$. The corresponding unconstrained mean is 3.00 with interval $(2.00, 4.01)$, all in assimilation units. The 95% intervals for the difference in marginal gain between 100 and 1,500 are positive under both fits, so the ordinary Gaussian analysis already supports diminishing gain for this pair of irradiances. Conditioning extends that ordering to the entire interval while reducing posterior uncertainty about marginal gain at high irradiance.

The upper endpoint also reveals the limits of a physiological interpretation. Only 4.6% of concave draws have a nonnegative slope at the highest measured irradiance, even though assimilation increases between the two highest observed irradiances. The inferred decline near the endpoint depends on the curvature restriction and stationary prior, and cannot by itself establish photoinhibition in this leaf.

The example illustrates how a stipulated curvature restriction changes derivative inference for one sequence, whose observation frame does not support conclusions about dark respiration or a species response. Likewise, the analysis does not establish an irradiance optimum independently of the shape assumption.

Quantity	Unconstrained	Concave
$A(100)$	2.46 (2.11, 2.80)	2.67 (2.38, 2.97)
$A(500)$	9.08 (8.38, 9.78)	8.92 (8.49, 9.38)
$A(1{,}500)$	12.08 (11.36, 12.80)	12.27 (11.86, 12.70)
$\frac{\mathrm{d}A}{\mathrm{d}q}(100)$	0.0361 (0.0315, 0.0407)	0.0357 (0.0316, 0.0399)
$\frac{\mathrm{d}A}{\mathrm{d}q}(1{,}500)$	0.0029 (-0.0099, 0.0155)	0.0019 (0.0010, 0.0028)
$\frac{\mathrm{d}A}{\mathrm{d}q}(100) - \frac{\mathrm{d}A}{\mathrm{d}q}(1{,}500)$	0.0332 (0.0197, 0.0467)	0.0338 (0.0295, 0.0381)
$A(1{,}500) - A(500)$	3.00 (2.00, 4.01)	3.35 (2.73, 3.99)

Table 3: Posterior means with pointwise 95% equal-tail credible intervals for the leaf light-response model in Equation (17), with arguments measured in μmol quanta $\mathrm{m}^{-2}\mathrm{s}^{-1}$. Response values and their differences have units μmol CO_2 $\mathrm{m}^{-2}\mathrm{s}^{-1}$, while derivatives and their difference have units mol CO_2 per mol quanta. Both columns use the same covariance and noise estimates, with the concave column additionally conditioning on nonpositive curvature throughout the observed range.

6 Discussion

This paper develops a common conditioning construction for positivity, monotonicity, and convexity in the Laplacian eigenfunction expansion. Its contribution is to express these whole-interval events within the HSGP model using its existing basis and kernel-derived Gaussian prior, including joint restrictions when a suitable common direction is available. The resulting representation supports joint inference for the curve and its derivatives and separates the definition of the conditioned posterior from the numerical evaluation of its threshold.

The approximation theory connects this finite construction to conditioning the underlying GP. For the individual constraints, the connection holds under the stated spectral assumptions as both padding and frequency cutoff increase, giving a basis for using finite expansions to approximate the corresponding constrained process. The joint construction is exact within the finite expansion, while the convergence theorem covers the individual restrictions.

The practical cost of the representation comes from evaluating a global threshold at each posterior-density calculation. Its maximizing location can change with the coefficients, creating kinks that matter for gradient-based sampling even when the marginal law is strongly log-concave. The threshold-error analysis controls the posterior change caused by conservative evaluation, while sampler accuracy and computational cost remain separate considerations. Developing samplers that exploit this geometry while accommodating the nonsmooth threshold is a useful direction for further work.

The mercury and leaf applications show that shape restrictions can substantially change derivative inference even when the fitted response curves remain similar. These changes matter because the derivatives determine the inferred mercury concentration and marginal carbon assimilation, respectively. Comparing the constrained and ordinary Gaussian posteriors makes the contribution of the shape assumption explicit in both the estimates and their uncertainty. The construction thus connects scientifically motivated restrictions to joint inference for the response and its derivatives, with the chosen constraint enforced throughout the interval.

References

Christian Agrell. Gaussian processes with linear operator inequality constraints. *Journal of Machine Learning Research*, 20(135):1–36, 2019.

Dimitri P. Bertsekas. Convex optimization algorithms, 2010. Supplementary Chapter 6 of Convex Optimization Theory. Version of October 8, 2010.

Michael Betancourt. A conceptual introduction to Hamiltonian Monte Carlo, 2017. Preprint, arXiv:1701.02434. Version 2, July 15, 2018.

Patrick Billingsley. *Convergence of Probability Measures*. Wiley, 2 edition, 1999. doi: 10.1002/9780470316962.

Bob Carpenter, Andrew Gelman, Matthew D. Hoffman, Daniel Lee, Ben Goodrich, Michael Betancourt, Marcus Brubaker, Jiqiang Guo, Peter Li, and Allen Riddell. Stan: A probabilistic programming language. *Journal of Statistical Software*, 76(1):1–32, 2017. doi: 10.18637/jss.v076.i01.

S. McKay Curtis and Sujit K. Ghosh. A variable selection approach to monotonic regression with Bernstein polynomials. *Journal of Applied Statistics*, 38(5):961–976, 2011. doi: 10.1080/02664761003692423.

John M. Danskin. The theory of max-min, with applications. *SIAM Journal on Applied Mathematics*, 14(4):641–664, 1966. doi: 10.1137/0114053.

DLMF. NIST Digital Library of Mathematical Functions. Release 1.2.8 of 2026-09-15, 2026. F. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, eds.

Oliver Y. Feng, Yining Chen, Qiyang Han, Raymond J. Carroll, and Richard J. Samworth. Nonparametric, tuning-free estimation of S-shaped functions. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 84(4):1324–1352, 2022. doi: 10.1111/rssb.12481.

Adam Gorm Hoffmann, Claus Thorn Ekstrøm, Benjamin Zeymer Christoffersen, and Andreas Kryger Jensen. Gaussian process regression for value-censored functional and longitudinal data. *Statistics in Medicine*, 44(20–22):e70277, 2025. doi: 10.1002/sim.70277.

Lauren A. Hannah and David B. Dunson. Bayesian nonparametric multivariate convex regression, 2011. Preprint, arXiv:1109.0322.

Matthew D. Hoffman and Andrew Gelman. The No-U-Turn sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15(47):1593–1623, 2014.

Ulla Holst, Ola Hössjer, Claes Björklund, Pär Ragnarson, and Hans Edner. Locally weighted least squares kernel regression and statistical evaluation of LIDAR measurements. *Environmetrics*, 7(4):401–416, 1996. doi: 10.1002/(SICI)1099-095X(199607)7:4<401::AID-ENV221>3.0.CO;2-D.

Andreas Kryger Jensen and Adam Gorm Hoffmann. Gaussian process differential ensembles for joint inference on curves, derivatives, and integrals, 2026. Preprint, arXiv:2606.23036.

Fredrik Johansson. Arb: Efficient arbitrary-precision midpoint-radius interval arithmetic. *IEEE Transactions on Computers*, 66(8):1281–1292, 2017. doi: 10.1109/TC.2017.2690633.

Olav Kallenberg. *Foundations of Modern Probability*. Springer, 3 edition, 2021. doi: 10.1007/978-3-030-61871-1.

Peter Lenk, Jangwon Lee, Dongu Han, Jichan Park, and Taeryon Choi. Hierarchical Bayesian spectral regression with shape constraints for multi-group data. *Computational Statistics & Data Analysis*, 200:108036, 2024. doi: 10.1016/j.csda.2024.108036.

Peter J. Lenk and Taeryon Choi. Bayesian analysis of shape-restricted functions using Gaussian process priors. *Statistica Sinica*, 27(1):43–69, 2017. doi: 10.5705/ss.202015.0096.

Lizhen Lin and David B. Dunson. Bayesian monotone regression using Gaussian process projection. *Biometrika*, 101(2):303–317, 2014. doi: 10.1093/biomet/ast063.

Andrés F. López-Lopera, François Bachoc, Nicolas Durrande, and Olivier Roustant. Finite-dimensional Gaussian approximation with linear inequality constraints. *SIAM/ASA Journal on Uncertainty Quantification*, 6(3):1224–1255, 2018. doi: 10.1137/17M1153157.

Andrés F. López-Lopera, François Bachoc, Nicolas Durrande, J. Rohmer, D. Idier, and Olivier Roustant. Approximating Gaussian process emulators with linear inequality constraints and noisy observations via MC and MCMC. In Bruno Tuffin and Pierre L'Ecuyer, editors, *Monte Carlo and Quasi-Monte Carlo Methods*, volume 324 of *Springer Proceedings in Mathematics & Statistics*, pages 363–381. Springer, Cham, 2020. doi: 10.1007/978-3-030-43465-6_18.

Hassan Maatouk and Xavier Bay. Gaussian process emulators for computer experiments with inequality constraints. *Mathematical Geosciences*, 49(5):557–582, 2017. doi: 10.1007/s11004-017-9673-2.

Hassan Maatouk, Didier Rullière, and Xavier Bay. Bayesian analysis of constrained Gaussian processes. *Bayesian Analysis*, 20(3):973–1002, 2025a. doi: 10.1214/24-BA1429.

Hassan Maatouk, Didier Rullière, and Xavier Bay. Large-scale constrained Gaussian processes for shape-restricted function estimation. *Statistics and Computing*, 35(1):7, 2025b. doi: 10.1007/s11222-024-10541-2.

Radford M. Neal. MCMC using Hamiltonian dynamics. In Steve Brooks, Andrew Gelman, Galin L. Jones, and Xiao-Li Meng, editors, *Handbook of Markov Chain Monte Carlo*, pages 113–162. Chapman & Hall/CRC, 2011. doi: 10.1201/b10905-6.

Brian Neelon and David B. Dunson. Bayesian isotonic regression and trend analysis. *Biometrics*, 60 (2):398–406, 2004. doi: 10.1111/j.0006-341X.2004.00184.x.

Carl Edward Rasmussen and Christopher K. I. Williams. *Gaussian Processes for Machine Learning*. Adaptive Computation and Machine Learning. MIT Press, 2006. ISBN 9780262182539.

Pallavi Ray, Debdeep Pati, and Anirban Bhattacharya. Efficient Bayesian shape-restricted function estimation with constrained Gaussian process priors. *Statistics and Computing*, 30(4):839–853, 2020. doi: 10.1007/s11222-020-09922-0.

Jaakko Riihimäki and Aki Vehtari. Gaussian processes with monotonicity information. In Yee Whye Teh and Mike Titterington, editors, *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics*, volume 9 of *Proceedings of Machine Learning Research*, pages 645–652. PMLR, 2010.

Gabriel Riutort-Mayol, Paul-Christian Bürkner, Michael R. Andersen, Arno Solin, and Aki Vehtari. Practical Hilbert space approximate Bayesian Gaussian processes for probabilistic programming. *Statistics and Computing*, 33(1):17, 2023. doi: 10.1007/s11222-022-10167-2.

Alistair Rogers, Kim S. Ely, and Shawn P. Serbin. Leaf photosynthetic parameters: Quantum yield, convexity, respiration, gross CO_2 assimilation rate and raw gas exchange data, Utqiagvik (Barrow), Alaska, 2016, 2019a. Dataset.

Alistair Rogers, Shawn P. Serbin, Kim S. Ely, and Stan D. Wullschleger. Terrestrial biosphere models may overestimate Arctic CO_2 assimilation if they do not account for decreased quantum yield and convexity at low temperature. *New Phytologist*, 223(1):167–179, 2019b. doi: 10.1111/nph.15750.

Adrien Saumard and Jon A. Wellner. Log-concavity and strong log-concavity: A review. *Statistics Surveys*, 8:45–114, 2014. doi: 10.1214/14-SS107.

Thomas S. Shively, Thomas W. Sager, and Stephen G. Walker. A Bayesian approach to non-parametric monotone function estimation. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 71(1):159–175, 2009. doi: 10.1111/j.1467-9868.2008.00677.x.

Arno Solin and Simo Särkkä. Hilbert space methods for reduced-rank Gaussian process regression. *Statistics and Computing*, 30(2):419–446, 2020. doi: 10.1007/s11222-019-09886-w.

A. W. van der Vaart and J. H. van Zanten. Reproducing kernel Hilbert spaces of Gaussian priors. In Bertrand Clarke and Subhashis Ghosal, editors, *Pushing the Limits of Contemporary Statistics: Contributions in Honor of Jayanta K. Ghosh*, volume 3 of *IMS Collections*, pages 200–222. Institute of Mathematical Statistics, 2008. doi: 10.1214/074921708000000156.

Ander Wilson, Jessica Tryner, Christian L'Orange, and John Volckens. Bayesian nonparametric monotone regression. *Environmetrics*, 31(8):e2642, 2020. doi: 10.1002/env.2642.

Chao Zhang, Jasper M. Everink, and Jakob Sauer Jørgensen. Efficient monotonic Gaussian processes via Randomize-then-Optimize. *Journal of Computational Physics*, 555:114787, 2026. doi: 10.1016/j.jcp.2026.114787.

A Proofs and supporting results

A.1 Gaussian conditioning and its specializations

Theorem 1 (Gaussian conditioning under a continuum sign restriction)**.** *Under the finite Gaussian model in Section 2, let* $\boldsymbol{a} = (a_0, \ldots, a_m)^T$ *denote the coefficient vector and fix* $r \in \{0, 1, 2\}$*. Suppose an invertible linear change of coordinates* $\boldsymbol{a} \mapsto (\boldsymbol{g}, \gamma) \in \mathbb{R}^m \times \mathbb{R}$ *gives* $f_{\boldsymbol{a}} = f_{\boldsymbol{g}} + \gamma h_r$*. Here* h_r *is a fixed function in the retained span, chosen to have a uniformly positive* r*th derivative on* I*, and* γ *is its scalar coefficient. The curve* $f_{\boldsymbol{g}}$ *depends linearly on the remaining coordinates* $\boldsymbol{g}$*. Assume these functions belong to* $C^r(I)$ *with* $\min_{x \in I} \frac{\mathrm{d}^r h_r}{\mathrm{d}x^r}(x) > 0$*. In these coordinates, the ordinary Gaussian posterior has the form*

$$\boldsymbol{g} \mid \boldsymbol{y} \sim N(\boldsymbol{m}_g, \boldsymbol{V}_g), \qquad \gamma \mid \boldsymbol{g}, \boldsymbol{y} \sim N(\mu(\boldsymbol{g}), s^2). \tag{4}$$

Here $\boldsymbol{V}_g$ *is positive definite,* μ *is affine, and* $s > 0$ *is constant in* $\boldsymbol{g}$*. The finite, attained threshold*

$$\tau_r(\boldsymbol{g}) = \max_{x \in I} \frac{-\frac{\mathrm{d}^r f_{\boldsymbol{g}}}{\mathrm{d}x^r}(x)}{\frac{\mathrm{d}^r h_r}{\mathrm{d}x^r}(x)} \tag{5}$$

satisfies $f_{\boldsymbol{a}} \in \mathcal{C}_r$ *exactly when* $\gamma \geq \tau_r(\boldsymbol{g})$*. This event has positive ordinary posterior probability and a null boundary in coefficient space. Its marginal density for* $\boldsymbol{g}$ *with respect to Lebesgue measure on* $\mathbb{R}^m$ *is*

$$p(\boldsymbol{g} \mid \boldsymbol{y}, \mathcal{C}_r) = \frac{p(\boldsymbol{g} \mid \boldsymbol{y}) \Phi^{\mathrm{c}}((\tau_r(\boldsymbol{g}) - \mu(\boldsymbol{g}))/s)}{P(f_{\boldsymbol{a}} \in \mathcal{C}_r \mid \boldsymbol{y})}. \tag{6}$$

Given this draw of $\boldsymbol{g}$*, draw* γ *from the conditional normal in Equation* (4) *truncated below at* $\tau_r(\boldsymbol{g})$ *and apply the inverse coefficient map. Equivalently, for an independent* $U \sim \mathrm{Uniform}(0, 1)$*,*

$$\gamma = \mu(\boldsymbol{g}) - s\Phi^{-1}(U\Phi^{\mathrm{c}}((\tau_r(\boldsymbol{g}) - \mu(\boldsymbol{g}))/s)). \tag{7}$$

This reconstructs the full conditioned coefficient law, under which $\min_I \frac{\mathrm{d}^r f_{\boldsymbol{a}}}{\mathrm{d}x^r} > 0$ *almost surely.*

Proof of Theorem 1. Fix r, I, the invertible map, and the ordinary posterior under the finite Gaussian model in Section 2 and the positive-direction hypotheses of Theorem 1, and write $b_r = \min_{x \in I} \frac{\mathrm{d}^r h_r}{\mathrm{d}x^r}(x) > 0$. The role of the positive direction is to turn every derivative inequality into a lower bound on the same scalar coefficient: division by $\frac{\mathrm{d}^r h_r}{\mathrm{d}x^r}$ gives the threshold equivalence, and continuity on compact I, together with $b_r > 0$, makes Equation (5) finite and attained. Since the invertible map preserves positive definiteness of the posterior covariance, Gaussian conditioning (Rasmussen and Williams, 2006, Section 2.2) gives Equation (4), including a positive Schur complement s^2.

Integrating the restricted joint Gaussian density over the scalar coefficient now leaves its conditional survival probability as a weight on $\boldsymbol{g}$, which proves Equation (6). This weight is strictly positive for every $\boldsymbol{g}$, so the conditioning event has positive probability. Given a draw from the weighted marginal, conditioning the Gaussian law of $\gamma \mid \boldsymbol{g}, \boldsymbol{y}$ on $\gamma \geq \tau_r(\boldsymbol{g})$ gives the stated truncated normal, whose survival-function inverse is Equation (7). Applying the inverse coefficient map therefore recovers the original coefficient law conditional on the sign restriction.

The minimum derivative is continuous in the coefficients, so coefficient vectors with a positive minimum lie in the event's interior and those with a negative minimum lie in its exterior. Conditional on $\boldsymbol{g}$, the nondegenerate normal assigns zero mass to equality at the threshold. At equality an attained maximizing location has zero derivative, whereas above it

$$\min_I \frac{\mathrm{d}^r f_{\boldsymbol{a}}}{\mathrm{d}x^r} \geq (\gamma - \tau_r(\boldsymbol{g}))b_r > 0.$$

At equality, an arbitrarily small increase or decrease in γ moves to either side of the event, identifying its coefficient boundary as precisely the threshold graph. Its zero Gaussian probability implies that the reconstructed law lies strictly above the graph almost surely, proving the strict-derivative assertion. □

Corollary 1 (Positivity in the Laplacian expansion)**.** *Under the finite Gaussian model in Section 2, suppose $m \geq 1$ and $I \subset (c-L, c+L)$. Set $h_0 = 1$, $\boldsymbol{g} = (a_1, \ldots, a_m)^T$, and $f_{\boldsymbol{g}} = \sum_{j=1}^m a_j \phi_{j,L}$. Then $f_{\boldsymbol{a}} = f_{\boldsymbol{g}} + a_0 h_0$, with inverse obtained by reinserting a_0.*

For $\theta(x) = \omega_1(x - c + L)$, $w(x) = \cos\theta(x)$, and $J = [w(x_+), w(x_-)] \subset (-1, 1)$, define $P_{0,\boldsymbol{g}}(w) = \sum_{j=1}^m j a_j T_j(w)$. The threshold is

$$\tau_0(\boldsymbol{g}) = -\min_{x \in I} f_{\boldsymbol{g}}(x). \tag{8}$$

Theorem 1 applies with scalar coefficient $\gamma = a_0$.

The threshold is attained at a physical endpoint or a location whose cosine coordinate is an interior real root in J of $P_{0,\boldsymbol{g}}$. This stationary polynomial has degree at most m when nonzero. All root multiplicities and reduced degrees are included. If the stationary polynomial vanishes identically, the curve $f_{\boldsymbol{g}}$ is constant and either endpoint suffices.

Proof of Corollary 1. Use the finite Gaussian model in Section 2 and its compact physical interval. Separating the intercept is an invertible permutation of the coefficients, with $\min_I h_0 = 1$, so Theorem 1 at order zero gives Equation (8) and the conditioned law. To locate the minimum over the whole interval, use the Chebyshev identities to write

$$f_{\boldsymbol{g}}(x) = \frac{\sqrt{1 - w(x)^2}}{\sqrt{L}} \sum_{j=1}^m a_j U_{j-1}(w(x)), \qquad \frac{\mathrm{d} f_{\boldsymbol{g}}}{\mathrm{d}x}(x) = \frac{\omega_1}{\sqrt{L}} P_{0,\boldsymbol{g}}(w(x)).$$

Because the cosine is strictly decreasing for $0 < \theta(x) < \pi$, interior stationary locations correspond exactly to interior roots of the displayed polynomial. The extreme value theorem and Fermat's theorem therefore give the endpoint and stationary candidates, with finitely many distinct roots, including repeated roots, when the polynomial is nonzero and has degree at most m. If it vanishes identically, the derivative is zero throughout I, so the mean value theorem gives constancy. Although the intercept is independent of the spectral coefficients in the prior, reconstruction uses its conditional posterior given $\boldsymbol{g}$ and the data. □

Corollary 2 (Monotonicity in the Laplacian expansion)**.** *Under the finite Gaussian model in Section 2, suppose $m \geq 4$ and $I \subset [c - L/2, c + L/2]$. Set $h_1 = -\phi_{2,L} - \phi_{4,L}/4$, $\boldsymbol{g} = (g_0, g_1, g_3, \ldots, g_m)^T$, and $f_{\boldsymbol{g}} = g_0 + \sum_{j=1}^m g_j \phi_{j,L}$, where $g_2 = 0$, $g_4 = a_4 - a_2/4$, and $g_j = a_j$ for $j \notin \{2, 4\}$. Then $f_{\boldsymbol{a}} = f_{\boldsymbol{g}} + (-a_2) h_1$, with inverse obtained by reinserting a_2 and setting $a_4 = g_4 + a_2/4$.*

For $\theta(x) = \omega_1(x - c + L)$, $w(x) = \cos\theta(x)$, and $J = [w(x_+), w(x_-)] \subset [-1/\sqrt{2}, 1/\sqrt{2}]$, define

$$P_{\boldsymbol{g}}(w) = \sum_{\substack{1 \leq j \leq m \\ j \neq 2}} j g_j T_j(w), \qquad D(w) = 1 + 4w^2 - 8w^4. \tag{9}$$

Since $D \geq 1$ on J, the threshold is

$$\tau_1(\boldsymbol{g}) = \max_{w \in J} \frac{-P_{\boldsymbol{g}}(w)}{D(w)}. \tag{10}$$

Theorem 1 applies with scalar coefficient $\gamma = -a_2$.

The threshold is attained at a physical endpoint or a location whose cosine coordinate is an interior real root in J of $\frac{\mathrm{d}P_{\boldsymbol{g}}}{\mathrm{d}w} D - P_{\boldsymbol{g}} \frac{\mathrm{d}D}{\mathrm{d}w}$. This stationary polynomial has degree at most $m + 3$

when nonzero. All root multiplicities and reduced degrees are included. If the stationary polynomial vanishes identically, the quotient $-P_{\boldsymbol{g}}/D$ is constant and either endpoint suffices. A nonzero constant stationary polynomial gives only endpoint candidates.

Proof of Corollary 2. Use the finite Gaussian model in Section 2 with the rank and closed padding interval in Corollary 2. Substitution of the inverse map proves the curve identity and invertibility, while the identity $T_j(w(x)) = \cos(j\theta(x))$ gives the derivative formulas

$$\frac{\mathrm{d}f_{\boldsymbol{g}}}{\mathrm{d}x}(x) = \frac{\omega_1}{\sqrt{L}} P_{\boldsymbol{g}}(w(x)), \qquad \frac{\mathrm{d}h_1}{\mathrm{d}x}(x) = \frac{\omega_1}{\sqrt{L}} D(w(x)).$$

The assumed padding confines $\theta(x)$ to $[\pi/4, 3\pi/4]$, where the cosine is strictly decreasing and $D(w) = 1 + 4w^2(1 - 2w^2) \geq 1$, including at the physical endpoints. The positive derivative bound therefore allows Theorem 1 to supply the conditioned law and strictness. Because the map transports the original Gaussian covariance, g_4 and a_2 retain their induced dependence.

Since $D > 0$ on compact J, the extreme value theorem and Fermat's theorem locate the maximum of $-P_{\boldsymbol{g}}/D$ at an endpoint or an interior stationary point. The quotient rule gives the stationary polynomial in the statement, whose degree is at most $m+3$ because $P_{\boldsymbol{g}}$ and D have degrees at most m and 4. If this polynomial vanishes identically, the quotient is constant by the mean value theorem and either endpoint suffices. Otherwise it has finitely many distinct roots, each a candidate when it lies in the interior of J, regardless of multiplicity. This also covers reduced degrees, including a nonzero constant stationary polynomial with no interior candidates. □

Corollary 3 (Convexity in the Laplacian expansion)**.** *Under the finite Gaussian model in Section 2, suppose $m \geq 2$ and $I \subset (c - L, c + L)$. Set $h_2 = -\phi_{1,L}$, $\boldsymbol{g} = (a_0, a_2, \ldots, a_m)^T$, and $f_{\boldsymbol{g}} = a_0 + \sum_{j=2}^{m} a_j \phi_{j,L}$. Then $f_{\boldsymbol{a}} = f_{\boldsymbol{g}} + (-a_1)h_2$, with inverse obtained by reinserting a_1.*

For $\theta(x) = \omega_1(x - c + L)$, $w(x) = \cos\theta(x)$, and $J = [w(x_+), w(x_-)] \subset (-1, 1)$, define the polynomial and threshold

$$Q_{\boldsymbol{g}}(w) = \sum_{j=2}^{m} j^2 a_j U_{j-1}(w), \qquad \tau_2(\boldsymbol{g}) = \max_{w \in J} Q_{\boldsymbol{g}}(w). \tag{11}$$

Theorem 1 applies with scalar coefficient $\gamma = -a_1$.

The threshold is attained at a physical endpoint or a location whose cosine coordinate is an interior real root in J of $\frac{\mathrm{d}Q_{\boldsymbol{g}}}{\mathrm{d}w}$. The polynomial $Q_{\boldsymbol{g}}$ has degree at most $m - 1$, so its stationary polynomial has degree at most $m - 2$ when nonzero. All root multiplicities and reduced degrees are included. If the stationary polynomial vanishes identically, $Q_{\boldsymbol{g}}$ is constant and either endpoint suffices. A nonzero constant stationary polynomial gives only endpoint candidates.

Proof of Corollary 3. Use the finite Gaussian model in Section 2 with the rank and open padding interval in Corollary 3. The coordinate map is an invertible signed permutation. Since $0 < \theta(x) < \pi$ on compact I,

$$\frac{\mathrm{d}^2 h_2}{\mathrm{d}x^2}(x) = \frac{\omega_1^2}{\sqrt{L}} \sin\theta(x) > 0, \qquad \frac{\mathrm{d}^2 f_{\boldsymbol{g}}}{\mathrm{d}x^2}(x) = -\frac{\omega_1^2}{\sqrt{L}} \sin\theta(x)\, Q_{\boldsymbol{g}}(w(x)).$$

The second formula uses $\sin(j\theta(x)) = \sin\theta(x)\, U_{j-1}(w(x))$, while compactness keeps the positive curvature in the first formula bounded away from zero. Thus Theorem 1 gives the claimed threshold and posterior under the stated open padding condition.

The extreme value theorem and Fermat's theorem locate the maximum of $Q_{\boldsymbol{g}}$ at an endpoint or an interior zero of $\frac{\mathrm{d}Q_{\boldsymbol{g}}}{\mathrm{d}w}$. The degree of $Q_{\boldsymbol{g}}$ is at most $m - 1$, so this derivative has degree at most

$m-2$ when nonzero. If it vanishes identically, $Q_{\boldsymbol{g}}$ is constant and either endpoint suffices. Otherwise its finitely many distinct interior roots enter the candidate set regardless of multiplicity, with none when the derivative is a nonzero constant. The same argument covers every reduced degree. □

Proposition 1 (Joint sign constraints)**.** *Under the finite Gaussian model in Section 2, let* $\boldsymbol{a} = (a_0, \ldots, a_m)^T$ *denote the coefficient vector and fix a nonempty* $A \subseteq \{0, 1, 2\}$ *and signs* $\varepsilon : A \to \{-1, 1\}$*, with event* $\mathcal{C}_A^\varepsilon$ *defined in Equation* (12)*. Suppose an invertible linear change of coordinates* $\boldsymbol{a} \mapsto (\boldsymbol{g}, \gamma) \in \mathbb{R}^m \times \mathbb{R}$ *gives* $f_{\boldsymbol{a}} = f_{\boldsymbol{g}} + \gamma h$*. Here* h *is a fixed function in the retained span, shared by all selected orders,* γ *is its scalar coefficient, and* $f_{\boldsymbol{g}}$ *depends linearly on the remaining coordinates* $\boldsymbol{g}$*. Assume* $h, f_{\boldsymbol{g}} \in C^{\max A}(I)$ *with* $\min_{x \in I} \varepsilon_r \frac{\mathrm{d}^r h}{\mathrm{d}x^r}(x) > 0$ *for every* $r \in A$*. Define*

$$\eta_r(x; \boldsymbol{g}) = \frac{-\varepsilon_r \frac{\mathrm{d}^r f_{\boldsymbol{g}}}{\mathrm{d}x^r}(x)}{\varepsilon_r \frac{\mathrm{d}^r h}{\mathrm{d}x^r}(x)}, \qquad \tau_A^\varepsilon(\boldsymbol{g}) = \max_{r \in A} \max_{x \in I} \eta_r(x; \boldsymbol{g}). \tag{13}$$

This finite, attained threshold satisfies $f_{\boldsymbol{a}} \in \mathcal{C}_A^\varepsilon$ *exactly when* $\gamma \geq \tau_A^\varepsilon(\boldsymbol{g})$*. The event has positive ordinary posterior probability and a null coefficient boundary. Its marginal density and reconstruction are Equations* (6) *and* (7)*, with* $\tau_r, \mathcal{C}_r$ *replaced by* $\tau_A^\varepsilon, \mathcal{C}_A^\varepsilon$ *and the Gaussian parameters in Equation* (4) *computed under this map. The conditioned law satisfies* $\min_{x \in I} \varepsilon_r \frac{\mathrm{d}^r f_{\boldsymbol{a}}}{\mathrm{d}x^r}(x) > 0$ *almost surely for every* $r \in A$*. The envelope, strong log-concavity, and conservative threshold-error conclusions extend under their respective hypotheses.*

Proof of Proposition 1. Fix the coefficient map and signed direction satisfying the hypotheses of Proposition 1. Dividing each signed derivative inequality by its positive direction derivative gives $\gamma \geq \eta_r(x; \boldsymbol{g})$, so their intersection gives Equation (13). Compactness of I, finiteness of A, and the denominator bounds make the maximum finite and attained. The Gaussian conditioning argument in the proof of Theorem 1 then gives positive probability, the marginal density, and reconstruction.

Above the threshold, every selected signed derivative has a strictly positive minimum. At equality, an attained maximizing pair gives a zero derivative, and perturbing γ crosses the event boundary. Thus the boundary is the threshold graph, which has zero conditional Gaussian probability for every $\boldsymbol{g}$, establishing almost-sure strictness. Each $\eta_r(x; \boldsymbol{g})$ is linear in $\boldsymbol{g}$, so the envelope and strong log-concavity arguments apply with the compact index set $A \times I$, and Proposition 2 applies to the same scalar truncation. □

A.2 Conservative threshold error

Proposition 2 (Posterior error from a conservative continuum threshold)**.** *Under Theorem 1, let* $\widetilde{\tau}_r$ *be measurable and satisfy* $\tau_r \leq \widetilde{\tau}_r \leq \tau_r + \delta$ *almost everywhere under the ordinary posterior of* $\boldsymbol{g}$*, for a fixed* $\delta \geq 0$*. Use the same upper threshold in marginal weighting and reconstruction. Let* $P_{\boldsymbol{y}}$ *be the ordinary coefficient posterior, and use* $\mathcal{C}_r$ *also for its coefficient preimage. Define* $\widetilde{\mathcal{C}}_r = \{\gamma \geq \widetilde{\tau}_r(\boldsymbol{g})\}$*. Both conditioned laws exist and, with* $\mathrm{TV}(P, Q) = \sup_A |P(A) - Q(A)|$*,*

$$\mathrm{TV}\big(P_{\boldsymbol{y}}(\cdot \mid \widetilde{\mathcal{C}}_r), P_{\boldsymbol{y}}(\cdot \mid \mathcal{C}_r)\big) = 1 - \frac{P_{\boldsymbol{y}}(\widetilde{\mathcal{C}}_r)}{P_{\boldsymbol{y}}(\mathcal{C}_r)} \leq \min\left(1, \frac{\delta}{s\sqrt{2\pi} P_{\boldsymbol{y}}(\mathcal{C}_r)}\right). \tag{15}$$

Every measurable summary of the reconstructed coefficients inherits this bound. The identity holds for any inner event of positive probability.

Proof of Proposition 2. Fix the ordinary Gaussian law, selected order, and almost-everywhere error condition of Proposition 2, modifying the numerical threshold on the exceptional null set if necessary. Finite thresholds and $s > 0$ give positive conditional tails, so both conditioning events have positive

probability. To compare the resulting laws, put $A = \mathcal{C}_r$ and $B = \widetilde{\mathcal{C}}_r$. If the omitted mass is zero, both conditioned laws agree. Otherwise the law on A is a mixture of the laws on B and $A \setminus B$, with weights $P_{\boldsymbol{y}}(B)/P_{\boldsymbol{y}}(A)$ and its complement. The components have disjoint supports, so total variation equals the second weight, establishing the identity for any positive-probability inner event.

For the threshold events considered here, conditional Gaussian integration identifies the omitted mass as

$$\mathrm{E}\left[\Phi((\widetilde{\tau}_r(\boldsymbol{g}) - \mu(\boldsymbol{g}))/s) - \Phi((\tau_r(\boldsymbol{g}) - \mu(\boldsymbol{g}))/s) \mid \boldsymbol{y}\right].$$

Since the conditional density is at most $1/(s\sqrt{2\pi})$ and the excluded interval has length at most δ, dividing this mass bound by the probability of the exact event proves Equation (15). The invertible coefficient map preserves total variation, while measurable summaries cannot increase it, proving the remaining assertion. □

A.3 Conditioned posterior approximation

Theorem 2 (Convergence of shape-constrained HSGP posteriors)**.** *Fix $r \in \{0, 1, 2\}$ under Assumptions 1 and 2, and consider finite models with $L \geq 2R$ and $m \geq 1$ for $r = 0$, $m \geq 4$ for $r = 1$, or $m \geq 2$ for $r = 2$. The underlying GP has a $C^r(I)$ version, and the finite and underlying shape events have positive probability and null boundaries in that space. Consequently, the posteriors conditional on the data and $\mathcal{C}_r$ exist, with finite moments of every polynomial order in $\|\cdot\|_{C^r}$. As $L \to \infty$ and $m/L \to \infty$,*

$$W_2(\Pi^{\mathcal{C}_r}_{m,L,\boldsymbol{y}}, \Pi^{\mathcal{C}_r}_{\boldsymbol{y}}) \longrightarrow 0. \tag{16}$$

This convergence also holds for the posterior summaries: means of $\frac{\mathrm{d}^j f}{\mathrm{d}x^j}$ converge uniformly on I for every $0 \leq j \leq r$, and covariances between $\frac{\mathrm{d}^{r_1} f}{\mathrm{d}x^{r_1}}(x)$ and $\frac{\mathrm{d}^{r_2} f}{\mathrm{d}x^{r_2}}(x')$ converge uniformly on $I \times I$ for every $0 \leq r_1, r_2 \leq r$. Under the conditioned underlying law, $\min_I \frac{\mathrm{d}^r f}{\mathrm{d}x^r} > 0$ almost surely.

Proof of Theorem 2. Fix r and any sequence of models satisfying the conditions and limits in Theorem 2. Write $F_{m,L}$ for a draw from the finite prior and $k_{m,L}(x, x') = \sum_{j=1}^m S_k(\omega_j)\phi_{j,L}(x)\phi_{j,L}(x')$ for the covariance of its spectral component. We first establish weak convergence of the priors in $C^r(I)$ and uniform norm moments. Positive shape probability and a boundary of prior probability zero will then allow passage to the conditioned posteriors, whose moment bounds give Wasserstein convergence and convergence of the stated summaries.

Covariance and increment bounds. Put $b = 2r + 2\zeta$. The two spectral weights in Assumption 2 also control the intermediate orders needed for covariance derivatives. For each integer $0 < d \leq 2r$, differentiating $\omega^d S_k(\omega)$ on $[0, 1]$ gives integrable terms because S_k is bounded there and its derivative is integrable. On $[1, \infty)$, write the same function as $\omega^{d-b}(\omega^b S_k(\omega))$. The multiplier and its derivative are bounded, so the product and its derivative are integrable. Thus, for $d = 0, \ldots, 2r$ and $d = b$, the weighted spectra are locally absolutely continuous and have themselves and their derivatives in $L^1(0, \infty)$, with the cases $d = 0$ and $d = b$ supplied directly by the assumption. Their finite values at zero and integrable derivatives also give boundedness. The derivative-covariance argument of Jensen and Hoffmann (2026, Theorem 2 and its proof) then gives, for $0 \leq r_1, r_2 \leq r$, with $d = r_1 + r_2$:

$$\sup_{x,x' \in I} \left| \frac{\partial^{r_1+r_2}(k_{m,L}(x, x') - k(x - x'))}{\partial x^{r_1} \partial (x')^{r_2}} \right| \leq \frac{b_d}{L} + \frac{1}{\pi} \int_{\pi m/(2L)}^{\infty} \omega^d S_k(\omega)\, d\omega, \tag{18}$$

where $b_d < \infty$ is independent of m and L. The cited theorem states stronger conditions on classical derivatives, but its proof bounds frequency-cell errors by integrals of absolute derivatives and the

reflected term by consecutive frequency differences. Both steps require only the absolute continuity, integrability, and boundedness verified above. In Equation (18), the first term vanishes as the padded boundary recedes, while the second vanishes as the frequency cutoff increases. The common intercept adds no covariance error.

For the frequency spacing $\Delta = \pi/(2L)$, with $\omega_j = j\Delta$, the same frequency-cell estimate bounds the spectral sums by

$$\Delta \sum_{j=1}^{m} (j\Delta)^d S_k(j\Delta) \le \|\omega^d S_k(\omega)\|_1 + \Delta \left\| \frac{\mathrm{d}(\omega^d S_k(\omega))}{\mathrm{d}\omega} \right\|_1, \qquad d \in \{2j : 0 \le j \le r\} \cup \{b\}.$$

Here the norms are over $[0, \infty)$. The right-hand side is uniformly bounded because $L \ge 2R$ gives $\Delta \le \pi/(4R)$. Taking $d = 0, 2, \ldots, 2r$ therefore bounds the derivative variances at x_-, with the intercept adding its fixed variance at order zero.

To control increments of the highest derivative, we use this estimate with $d = b = 2r + 2\zeta$. Each differentiated basis function is a sine or cosine with amplitude $L^{-1/2}\omega_j^r$, so $|\sin u - \sin v| \le 2^{1-\zeta}|u-v|^\zeta$ and the corresponding cosine inequality give

$$\left| \frac{\mathrm{d}^r \phi_{j,L}}{\mathrm{d}x^r}(x) - \frac{\mathrm{d}^r \phi_{j,L}}{\mathrm{d}x^r}(x') \right| \le \frac{2^{1-\zeta}}{\sqrt{L}} \omega_j^{r+\zeta} |x - x'|^\zeta.$$

By independence of the Gaussian coefficients, the variance of the rth derivative increment is the sum of the squared basis increments above weighted by $S_k(\omega_j)$. The intercept cancels in increments when $r = 0$ and contributes no derivative when $r \ge 1$. Consequently,

$$\mathrm{Var}\left(\frac{\mathrm{d}^r F_{m,L}}{\mathrm{d}x^r}(x) - \frac{\mathrm{d}^r F_{m,L}}{\mathrm{d}x^r}(x') \right) \le C_\zeta |x - x'|^{2\zeta} \Delta \sum_{j=1}^{m} \omega_j^{2r+2\zeta} S_k(\omega_j),$$

where $C_\zeta = 2^{3-2\zeta}/\pi$ follows from $1/L = 2\Delta/\pi$. The frequency-cell estimate at $d = b$ bounds the spectral sum on the right uniformly in m, L, giving the increment bound $A|x - x'|^{2\zeta}$ with A independent of rank and padding. Fourier inversion gives the same bound for the underlying mean-square rth derivative, after increasing A if needed. Gaussian moments then verify the Kolmogorov continuity criterion (Kallenberg, 2021) with any exponent p satisfying $p\zeta > 1$. For $r = 0$, this gives a continuous version of W. For $r \ge 1$, integrating the continuous highest derivative, with the lower mean-square derivative values at x_- as integration constants, recovers a version of W by the mean-square fundamental theorem of calculus. Adding the intercept therefore gives a $C^r(I)$ version of F.

Tightness and norm moments. We next control both the size and the continuity of the sample paths uniformly over the finite models. These bounds establish tightness in $C^r(I)$ and provide the norm moments needed later for Wasserstein convergence and convergence of posterior summaries. Let H denote either $\frac{\mathrm{d}^r F_{m,L}}{\mathrm{d}x^r}$ or $\frac{\mathrm{d}^r F}{\mathrm{d}x^r}$, and choose $p > \max(4, 1/\zeta)$ and a Hölder exponent $0 < \kappa < \zeta - 1/p$. For $j \ge 0$, let M_j be the largest absolute increment of H between neighboring points on the mesh dividing I into 2^j equal cells. The Gaussian increment bound gives $\mathrm{E}[M_j^p]^{1/p} \le C_p 2^{-(\zeta - 1/p)j}$ uniformly over the models, because the pth power of the maximum is bounded by the sum of the pth powers of the 2^j increments. Approximating arbitrary $x, x' \in I$ by points on successively finer meshes expresses $H(x) - H(x')$ through adjacent increments. This dyadic chaining argument bounds the Hölder seminorm by

$$[H]_\kappa = \sup_{\substack{x, x' \in I \\ x \ne x'}} \frac{|H(x) - H(x')|}{|x - x'|^\kappa} \le C_{I,\kappa} \sum_{j=0}^{\infty} 2^{\kappa j} M_j.$$

Here $C_{I,\kappa}$ depends only on I and κ. Minkowski's inequality therefore bounds $\mathrm{E}[[H]_\kappa^p]^{1/p}$ by a constant times $\sum_{j\geq 0} 2^{-(\zeta-1/p-\kappa)j}$. Our choice of κ makes this series converge, giving a uniform pth moment bound for the seminorm.

The seminorm and the value $H(x_-)$ together bound the highest derivative everywhere on I. When $r \geq 1$, repeated integration, starting from the lower derivative values at x_-, then bounds every lower derivative and the curve itself. For $r = 0$, the first bound already controls the curve. The uniformly bounded Gaussian moments of these endpoint values consequently give

$$\sup_{m,L} \mathrm{E}[\|F_{m,L}\|_{C^r}^p] + \mathrm{E}[\|F\|_{C^r}^p] < \infty \tag{19}$$

over the permitted ranks and padding. Since p can be arbitrarily large, every polynomial norm moment is uniformly bounded.

Bounds on derivative values of orders $0, \ldots, r$ at x_- and on the highest derivative's Hölder seminorm define relatively compact subsets of $C^r(I)$ by Arzelà–Ascoli. The moment bounds and Markov's inequality allow these bounds to be chosen so that the finite priors assign arbitrarily high probability to such a set, uniformly over the models. This proves tightness. To identify the limit, the common mean and Equation (18) give convergence of the joint Gaussian laws of every finite collection of derivative values. Such evaluations on a countable dense subset of I determine a $C^r(I)$ law, so every subsequential limit is the underlying prior. The finite priors therefore converge weakly to that law.

Positive shape probability and a null boundary. The boundary of $\mathcal{C}_r$ in $C^r(I)$ consists exactly of curves with $\min_I \frac{\mathrm{d}^r f}{\mathrm{d}x^r} = 0$. Indeed, this minimum is continuous in the C^r norm, and adding or subtracting a small multiple of $(x-c)^r/r!$ moves any such curve to either side of the event. We therefore need positive probability of nonnegativity and zero probability of equality. For each order, these follow by separating one Gaussian coordinate whose contribution to the selected derivative is positive throughout I.

For $r = 0$, the independent intercept gives

$$\min_{x\in I} F(x) = a_0 + \min_{x\in I} W(x).$$

Conditional on W, nonnegativity therefore imposes a finite lower threshold on the nondegenerate Gaussian a_0, giving positive conditional probability and zero conditional probability of equality.

For $r \in \{1, 2\}$, the intercept contributes no derivative. We instead seek a decomposition of W into a Gaussian coefficient multiplying a deterministic function whose rth derivative is uniformly positive, together with an independent Gaussian process. This reduces the shape restriction to a scalar threshold, as in the finite model. The reproducing kernel Hilbert space (RKHS) $\mathcal{H}$ of W in $C^r(I)$ identifies the deterministic directions that can be separated in this way.

To construct a candidate direction, choose $0 < \omega_- < \omega_+ < \pi/(2R)$ and set

$$v_r(x) = \int_{\omega_-}^{\omega_+} S_k(\omega) \cos(\omega(x-c) - r\pi/2)\, d\omega.$$

Taking r derivatives removes the phase shift, leaving the cosine factor $\cos(\omega(x-c))$. Because $|x-c| \leq R$ and $\omega_+ R < \pi/2$, this factor is uniformly positive on the selected frequency band and throughout I. Thus

$$\frac{\mathrm{d}^r v_r}{\mathrm{d}x^r}(x) \geq \cos(\omega_+ R) \int_{\omega_-}^{\omega_+} \omega^r S_k(\omega)\, d\omega > 0.$$

We next verify that this direction belongs to $\mathcal{H}$, which justifies separating a Gaussian coefficient along it. The Fourier expression for the full-line RKHS norm gives squared norm $\pi \int_{\omega_-}^{\omega_+} S_k(\omega)\, d\omega < \infty$ for v_r, and restricting the function to I cannot increase that norm. The RKHS obtained from the covariance on I agrees with the RKHS of W as a $C^r(I)$-valued Gaussian random element by van der Vaart and van Zanten (2008, Lemma 8.1), applied to the continuous, injective embedding into $C(I)$. The norm calculation therefore establishes $v_r \in \mathcal{H}$ in the function space used in this proof.

Since v_r is nonzero, we can normalize it by its RKHS norm. The Gaussian RKHS isometry (van der Vaart and van Zanten, 2008, Lemma 2.3) then supplies a standard normal variable ξ, jointly Gaussian with W, such that $\mathrm{E}[\xi W(x)] = v_r(x)/\|v_r\|_{\mathcal{H}}$ for every $x \in I$. Define the remaining process V by

$$W = \xi \frac{v_r}{\|v_r\|_{\mathcal{H}}} + V.$$

Subtracting this component preserves $C^r(I)$ paths, and $\mathrm{Var}(\xi) = 1$ gives $\mathrm{Cov}(\xi, V(x)) = 0$ for every $x \in I$. Joint Gaussianity therefore makes ξ independent of the whole process V.

Conditional on V, only the scalar coefficient ξ remains random in the derivative constraint. Dividing each derivative inequality by the uniformly positive derivative of v_r gives

$$\{F \in \mathcal{C}_r\} = \left\{ \xi \geq \|v_r\|_{\mathcal{H}} \max_{x \in I} \frac{-\frac{\mathrm{d}^r V}{\mathrm{d}x^r}(x)}{\frac{\mathrm{d}^r v_r}{\mathrm{d}x^r}(x)} \right\}.$$

For each path of V the threshold is finite, so the conditional normal tail has positive probability and no atom at equality. Equality is precisely the zero-minimum event identified above, so the underlying shape event has positive probability and a null boundary at all three orders. The finite laws have the same properties by Corollaries 1 to 3, applied with likelihood one. In both cases, conditioning on nonnegativity therefore gives a strictly positive minimum selected derivative almost surely.

Likelihood and shape conditioning. Write $\mathcal{L}_{\boldsymbol{y}}(f)$ for the Gaussian density at $\boldsymbol{y}$ with mean $(f(x_1), \ldots, f(x_n))^T$ and covariance $\boldsymbol{\Sigma}$, taking $\mathcal{L}_{\boldsymbol{y}}(f) = 1$ when $n = 0$. This likelihood is bounded, positive, and continuous on $C^r(I)$. For any bounded continuous test function $\psi : C^r(I) \to \mathbb{R}$, the map

$$f \longmapsto \psi(f)\mathcal{L}_{\boldsymbol{y}}(f)\mathbf{1}_{\mathcal{C}_r}(f)$$

is bounded and continuous outside $\partial\mathcal{C}_r$, where $\mathbf{1}_{\mathcal{C}_r}(f)$ is one if $f \in \mathcal{C}_r$ and zero otherwise. The preceding argument shows that this boundary has zero probability under the underlying prior, so the Portmanteau theorem (Billingsley, 1999, Section 2) gives convergence of the integrals of this map under the finite priors to its integral under the underlying prior. Taking $\psi \equiv 1$ shows that the normalizing constants converge to a positive limit. Dividing by them gives convergence of posterior expectations of every bounded continuous ψ, which is conditioned weak convergence. The posterior densities relative to the priors are uniformly bounded along the sequence by the likelihood supremum divided by these constants, so Equation (19) also holds for the conditioned laws at every fixed polynomial order. Likelihood weighting preserves the zero boundary probability, giving the strictness assertion.

Wasserstein convergence and posterior summaries. We now strengthen conditioned weak convergence to convergence in Wasserstein distance. Index the chosen approximation sequence by ℓ, with ranks m_ℓ and padding L_ℓ. In this paragraph, write F_ℓ and F for draws from the finite and underlying conditioned posteriors, respectively. Since $C^r(I)$ is separable and complete,

the Skorokhod representation theorem (Billingsley, 1999, Theorem 6.7) allows these draws to be constructed on a common probability space, preserving their posterior laws, so that $\|F_\ell - F\|_{C^r} \to 0$ almost surely. The uniform fourth-moment bounds established above make the squared norm differences uniformly integrable, so their expectations also converge to zero. Wasserstein distance takes the infimum over all couplings, and this particular coupling therefore gives

$$W_2^2(\Pi^{\mathcal{C}_r}_{m_\ell, L_\ell, \boldsymbol{y}}, \Pi^{\mathcal{C}_r}_{\boldsymbol{y}}) \le \mathrm{E}[\|F_\ell - F\|^2_{C^r}] \longrightarrow 0,$$

proving Equation (16).

The same coupling controls posterior summaries uniformly in their input locations because the C^r norm bounds every derivative through order r over the whole interval. For each such order, the uniform difference of derivative means is at most $\mathrm{E}[\|F_\ell - F\|_{C^r}]$, which tends to zero by Cauchy–Schwarz. For any two derivative orders through r, expand the difference of their products by changing one factor at a time. The resulting difference of mixed second moments is bounded, uniformly on $I \times I$, by

$$\mathrm{E}[\|F_\ell - F\|_{C^r}(\|F_\ell\|_{C^r} + \|F\|_{C^r})].$$

Cauchy–Schwarz bounds this expectation by the product of $\mathrm{E}[\|F_\ell - F\|^2_{C^r}]^{1/2}$, which tends to zero, and $\mathrm{E}[(\|F_\ell\|_{C^r} + \|F\|_{C^r})^2]^{1/2}$, which remains bounded. Thus the mixed second moments converge uniformly. Subtracting the uniformly convergent products of means proves covariance convergence. Since the sequence was arbitrary, all these conclusions hold in the full stated limiting regime. □

A.4 Signed combinations in the Laplacian expansion

Corollary 4 (Signed combinations and their reductions)**.** *Under the finite Gaussian model in Section 2, let $A \subseteq \{0,1,2\}$ be nonempty and fix signs $\varepsilon_r \in \{-1,1\}$ for $r \in A$. For $\theta(x) = \omega_1(x - c + L)$, $w(x) = \cos\theta(x)$, and $J = [w(x_+), w(x_-)] \subset (-1,1)$, define, for a coefficient array $\boldsymbol{b} = (b_0, \dots, b_m)^T$,*

$$H_{0,\boldsymbol{b}}(w) = \sqrt{L}\, b_0 + \sqrt{1-w^2} \sum_{j=1}^{m} b_j U_{j-1}(w),$$

$$H_{1,\boldsymbol{b}}(w) = \sum_{j=1}^{m} j b_j T_j(w), \qquad H_{2,\boldsymbol{b}}(w) = -\sum_{j=1}^{m} j^2 b_j U_{j-1}(w).$$

Suppose a fixed $\boldsymbol{d} = (d_0, \dots, d_m)^T$ satisfies $\min_{w \in J} \varepsilon_r H_{r,\boldsymbol{d}}(w) > 0$ for every $r \in A$. Set $h = d_0 + \sum_{j=1}^m d_j \phi_{j,L}$ and choose $j_* \in \{0, \dots, m\}$ with $d_{j_*} \neq 0$. The coordinates

$$\gamma = \frac{a_{j_*}}{d_{j_*}}, \qquad g_j = a_j - d_j\gamma \quad (j \neq j_*), \qquad g_{j_*} = 0$$

give $\boldsymbol{g} = (g_j)_{j \neq j_*} \in \mathbb{R}^m$ and $f_{\boldsymbol{g}} = g_0 + \sum_{j=1}^m g_j \phi_{j,L}$. Then $f_{\boldsymbol{a}} = f_{\boldsymbol{g}} + \gamma h$, with inverse $a_j = g_j + d_j\gamma$ for $0 \le j \le m$. In expressions involving $H_{r,\boldsymbol{g}}$, the zero coordinate g_{j_*} is included in the coefficient array.

With η_r from Equation (13)*, define*

$$\eta_r(x; \boldsymbol{g}) = -\frac{H_{r,\boldsymbol{g}}(w(x))}{H_{r,\boldsymbol{d}}(w(x))}, \qquad M_r(\boldsymbol{g}) = \max_{x \in I} \eta_r(x; \boldsymbol{g}), \qquad \eta_r^{\pm}(\boldsymbol{g}) = \eta_r(x_\pm; \boldsymbol{g}).$$

For adjacent selected orders $r, r+1 \in A$, let $E_r(\boldsymbol{g}) = \eta_r^-(\boldsymbol{g})$ when $\varepsilon_r = \varepsilon_{r+1}$ and $E_r(\boldsymbol{g}) = \eta_r^+(\boldsymbol{g})$ otherwise. Suppressing the argument $\boldsymbol{g}$ on the right, the threshold is

$$\tau_A^{\varepsilon}(\boldsymbol{g}) = \begin{cases} M_r, & A = \{r\}, \\ \max\{M_1, E_0\}, & A = \{0,1\}, \\ \max\{M_2, M_0\}, & A = \{0,2\},\ \varepsilon_0 = \varepsilon_2, \\ \max\{M_2, \eta_0^-, \eta_0^+\}, & A = \{0,2\},\ \varepsilon_0 \neq \varepsilon_2, \\ \max\{M_2, E_1\}, & A = \{1,2\}, \\ \max\{M_2, E_1, E_0\}, & A = \{0,1,2\}. \end{cases}$$

Proposition 1 applies with scalar coefficient $\gamma = a_{j_*}/d_{j_*}$. Thus $f_{\boldsymbol{a}} \in \mathcal{C}_A^{\varepsilon}$ exactly when $\gamma \geq \tau_A^{\varepsilon}(\boldsymbol{g})$.

Each maximum M_r is attained at a physical endpoint or a location whose cosine coordinate is an interior root in J of

$$H_{r,\boldsymbol{d}}(w)\frac{\mathrm{d}H_{r,\boldsymbol{g}}}{\mathrm{d}w}(w) - H_{r,\boldsymbol{g}}(w)\frac{\mathrm{d}H_{r,\boldsymbol{d}}}{\mathrm{d}w}(w) = 0.$$

For $r = 1, 2$, the left side is a polynomial. All root multiplicities and reduced degrees are included. If the left side vanishes identically, the quotient is constant and either endpoint suffices.

These formulas cover the 26 nonempty combinations without opposing signs, with positive ordinary posterior probability whenever the stated direction condition holds for the selected combination. The remaining 37 selections contain opposing signs. At orders zero, one, and two these force, respectively, a zero, constant, or affine function, subject to any further restrictions. In the finite expansion, affine curves are constant, so all 37 events have ordinary posterior probability zero.

Proof. Fix the selected orders, signs, and direction coefficients satisfying Corollary 4. The inverse map follows by substitution, so the coordinates preserve the full coefficient space. The Chebyshev identities in Definition 1 give

$$\begin{aligned} f_{\boldsymbol{b}}(x) &= \frac{H_{0,\boldsymbol{b}}(w(x))}{\sqrt{L}}, \\ \frac{\mathrm{d}f_{\boldsymbol{b}}}{\mathrm{d}x}(x) &= \frac{\omega_1}{\sqrt{L}} H_{1,\boldsymbol{b}}(w(x)), \\ \frac{\mathrm{d}^2 f_{\boldsymbol{b}}}{\mathrm{d}x^2}(x) &= \frac{\omega_1^2 \sin\theta(x)}{\sqrt{L}} H_{2,\boldsymbol{b}}(w(x)). \end{aligned}$$

Since $\sin\theta$ is positive on compact I, the assumed conditions on $H_{r,\boldsymbol{d}}$ give the uniform signed derivative bounds required by Proposition 1. The positive factors cancel from the ratios, yielding the displayed expression for η_r and the conditioned posterior.

A signed slope restriction makes the minimum of $\varepsilon_0 f_{\boldsymbol{a}}$ occur at x_- when $\varepsilon_0 = \varepsilon_1$, and at x_+ otherwise. This gives E_0. Applying the same argument to the signed slope $\varepsilon_1 \frac{\mathrm{d}f_{\boldsymbol{a}}}{\mathrm{d}x}$ under the curvature restriction gives E_1. When $\varepsilon_0 \neq \varepsilon_2$, the signed curve $\varepsilon_0 f_{\boldsymbol{a}}$ is concave, so its nonnegativity is equivalent to nonnegative endpoint values by the chord inequality. When the signs agree, the signed curve is convex and its minimum may be interior, so the value restriction retains M_0. Together with the continuum slope and curvature bounds M_1 and M_2, these equivalences give every displayed threshold, because each pointwise restriction is $\gamma \geq \eta_r(x; \boldsymbol{g})$.

The cosine coordinate is strictly decreasing on I, and the signed denominators stay positive on compact J. The extreme value theorem, Fermat's theorem, and the quotient rule therefore give the stated endpoint and stationary candidates. The functions $H_{r,\boldsymbol{b}}$ are analytic on $(-1, 1)$, so a nonzero

stationary expression has only finitely many roots on J. If it vanishes identically, the mean value theorem gives constancy of the quotient.

Choosing no restriction or one of two signs at each order gives $3^3-1=26$ nonempty combinations: six singles, twelve pairs, and eight triples. Of the $2^6-1=63$ nonempty selections from the six constraints, the other 37 contain opposing signs, which force the corresponding derivative to vanish identically. This gives zero, a constant, or an affine function. Analyticity extends any affine identity on I to the padded interval, whose two endpoint values both equal a_0, so the affine curve is constant. Linear independence of the sine basis gives $a_1=\cdots=a_m=0$, with $a_0=0$ as well when both value signs are imposed. These proper linear subspaces have zero probability under the nondegenerate Gaussian posterior. □

Table 4: Implementing the 20 joint combinations without opposing constraints under Corollary 4. P and N denote nonnegativity and nonpositivity, I and D denote nondecreasing and nonincreasing curves, and CX and CV denote convexity and concavity. Choose fixed direction coefficients $\boldsymbol{d}$ so that every signed profile in the second column is strictly positive throughout J. For a pivot $d_{j_*}\neq 0$, use $\gamma=a_{j_*}/d_{j_*}$ and $g_j=a_j-d_j\gamma$, so that $g_{j_*}=0$ and $a_j=g_j+d_j\gamma$ for $0\le j\le m$. Every displayed coefficient bound in a row must hold, giving $\gamma\ge\tau_A^{\varepsilon}(\boldsymbol{g})$ by taking the largest displayed right-hand side. With arguments $\boldsymbol{g}$ suppressed, $M_r=\max_{x\in I}\eta_r(x;\boldsymbol{g})$ requires an interval maximum, evaluated at the endpoint and stationary candidates in Corollary 4, whereas $\eta_r^{\pm}=\eta_r(x_{\pm};\boldsymbol{g})$ requires only evaluation at the left endpoint x_- or right endpoint x_+. All ratios $\eta_r(x;\boldsymbol{g})=-H_{r,\boldsymbol{g}}(w(x))/H_{r,\boldsymbol{d}}(w(x))$ use the same direction and coefficient map within a row. A dash indicates no separate endpoint bound.

Combination	Direction condition on J	Continuum bound	Endpoint bound
Pairs of constraints			
P, I	$H_{0,\boldsymbol{d}}, H_{1,\boldsymbol{d}}>0$	$\gamma\ge M_1$	$\gamma\ge\eta_0^-$
P, D	$H_{0,\boldsymbol{d}}, -H_{1,\boldsymbol{d}}>0$	$\gamma\ge M_1$	$\gamma\ge\eta_0^+$
N, I	$-H_{0,\boldsymbol{d}}, H_{1,\boldsymbol{d}}>0$	$\gamma\ge M_1$	$\gamma\ge\eta_0^+$
N, D	$-H_{0,\boldsymbol{d}}, -H_{1,\boldsymbol{d}}>0$	$\gamma\ge M_1$	$\gamma\ge\eta_0^-$
P, CX	$H_{0,\boldsymbol{d}}, H_{2,\boldsymbol{d}}>0$	$\gamma\ge\max\{M_2,M_0\}$	—
P, CV	$H_{0,\boldsymbol{d}}, -H_{2,\boldsymbol{d}}>0$	$\gamma\ge M_2$	$\gamma\ge\max\{\eta_0^-,\eta_0^+\}$
N, CX	$-H_{0,\boldsymbol{d}}, H_{2,\boldsymbol{d}}>0$	$\gamma\ge M_2$	$\gamma\ge\max\{\eta_0^-,\eta_0^+\}$
N, CV	$-H_{0,\boldsymbol{d}}, -H_{2,\boldsymbol{d}}>0$	$\gamma\ge\max\{M_2,M_0\}$	—
I, CX	$H_{1,\boldsymbol{d}}, H_{2,\boldsymbol{d}}>0$	$\gamma\ge M_2$	$\gamma\ge\eta_1^-$
I, CV	$H_{1,\boldsymbol{d}}, -H_{2,\boldsymbol{d}}>0$	$\gamma\ge M_2$	$\gamma\ge\eta_1^+$
D, CX	$-H_{1,\boldsymbol{d}}, H_{2,\boldsymbol{d}}>0$	$\gamma\ge M_2$	$\gamma\ge\eta_1^+$
D, CV	$-H_{1,\boldsymbol{d}}, -H_{2,\boldsymbol{d}}>0$	$\gamma\ge M_2$	$\gamma\ge\eta_1^-$
Triples of constraints			
P, I, CX	$H_{0,\boldsymbol{d}}, H_{1,\boldsymbol{d}}, H_{2,\boldsymbol{d}}>0$	$\gamma\ge M_2$	$\gamma\ge\max\{\eta_1^-,\eta_0^-\}$
P, I, CV	$H_{0,\boldsymbol{d}}, H_{1,\boldsymbol{d}}, -H_{2,\boldsymbol{d}}>0$	$\gamma\ge M_2$	$\gamma\ge\max\{\eta_1^+,\eta_0^-\}$
P, D, CX	$H_{0,\boldsymbol{d}}, -H_{1,\boldsymbol{d}}, H_{2,\boldsymbol{d}}>0$	$\gamma\ge M_2$	$\gamma\ge\max\{\eta_1^+,\eta_0^+\}$
P, D, CV	$H_{0,\boldsymbol{d}}, -H_{1,\boldsymbol{d}}, -H_{2,\boldsymbol{d}}>0$	$\gamma\ge M_2$	$\gamma\ge\max\{\eta_1^-,\eta_0^+\}$
N, I, CX	$-H_{0,\boldsymbol{d}}, H_{1,\boldsymbol{d}}, H_{2,\boldsymbol{d}}>0$	$\gamma\ge M_2$	$\gamma\ge\max\{\eta_1^-,\eta_0^+\}$
N, I, CV	$-H_{0,\boldsymbol{d}}, H_{1,\boldsymbol{d}}, -H_{2,\boldsymbol{d}}>0$	$\gamma\ge M_2$	$\gamma\ge\max\{\eta_1^+,\eta_0^+\}$
N, D, CX	$-H_{0,\boldsymbol{d}}, -H_{1,\boldsymbol{d}}, H_{2,\boldsymbol{d}}>0$	$\gamma\ge M_2$	$\gamma\ge\max\{\eta_1^+,\eta_0^-\}$
N, D, CV	$-H_{0,\boldsymbol{d}}, -H_{1,\boldsymbol{d}}, -H_{2,\boldsymbol{d}}>0$	$\gamma\ge M_2$	$\gamma\ge\max\{\eta_1^-,\eta_0^-\}$

A.5 Whitened posterior geometry

Proposition 3 (Strong log-concavity of the marginal posterior)**.** *Under Theorem 1, let* $\boldsymbol{B}\boldsymbol{B}^T = \boldsymbol{V}_g$ *with* $\boldsymbol{B}$ *invertible and put* $\boldsymbol{g} = \boldsymbol{m}_g + \boldsymbol{B}\boldsymbol{z}$*. The induced density* q_r *of* $\boldsymbol{z}$ *is* 1*-strongly log-concave. That is,* $\log q_r(\boldsymbol{z}) + \|\boldsymbol{z}\|^2/2$ *is concave. It has a unique mode and finite moments of every polynomial order.*

Proof. Fix the Gaussian posterior, selected order, and invertible whitening factor in Proposition 3. Since τ_r is a maximum of linear functions, it is convex. Subtracting the affine conditional mean and dividing by $s > 0$ therefore makes $\alpha_r = (\tau_r - \mu)/s$ convex in the whitened coordinates. Under the ordinary posterior, $\eta = (\gamma - \mu(\boldsymbol{g}))/s$ is standard normal and independent of $\boldsymbol{z}$. The joint standard normal density in $(\boldsymbol{z}, \eta)$ restricted to the convex epigraph $\eta \geq \alpha_r(\boldsymbol{z})$ is 1-strongly log-concave and has positive normalizer by Theorem 1. Preservation under marginalization (Saumard and Wellner, 2014, Theorem 3.8) gives the same constant for q_r. To obtain the moment and mode conclusions, use the survival factor's continuity, positivity, and upper bound of one: q_r is continuous and bounded above by the standard normal density divided by the positive event probability. This bound gives all polynomial moments and convergence of the density to zero at infinity, so continuity implies attainment of a mode, which strong log-concavity makes unique. □

B Numerical evaluation

The posterior construction requires three numerical operations. We first prepare the ordinary Gaussian posterior, then evaluate the shape threshold whenever the sampler proposes a coefficient vector, and finally draw the omitted coefficient conditionally on each retained vector. The Gaussian preparation is performed once, whereas threshold evaluation is repeated throughout sampling. The main numerical issue is therefore to bound a maximum over the entire constraint interval reliably and to use the same bound in the marginal density and conditional reconstruction.

B.1 Gaussian preparation and the common algorithm

The preparation separates the coefficients that will be sampled from the scalar that will be integrated out. Let $\boldsymbol{A}$ be the invertible linear map from the original coefficients to $(\boldsymbol{g}, \gamma)$ under the applicable conditions in Corollaries 1 to 3, or the common map in Corollary 4 for a joint restriction. The ordinary posterior then has mean $\boldsymbol{A}\boldsymbol{m}$ and covariance $\boldsymbol{A}\boldsymbol{V}\boldsymbol{A}^T$. Denote the mean blocks by $\boldsymbol{m}_g$ and m_γ, the marginal covariance of $\boldsymbol{g}$ by $\boldsymbol{V}_g$, the variance of γ by v_γ, and their cross-covariance by $\boldsymbol{v} = \mathrm{Cov}(\boldsymbol{g}, \gamma \mid \boldsymbol{y})$. Gaussian conditioning gives

$$\mu(\boldsymbol{g}) = m_\gamma + \boldsymbol{v}^T \boldsymbol{V}_g^{-1}(\boldsymbol{g} - \boldsymbol{m}_g), \qquad s^2 = v_\gamma - \boldsymbol{v}^T \boldsymbol{V}_g^{-1} \boldsymbol{v}. \tag{20}$$

These formulas give the conditional mean and variance of γ for any value of $\boldsymbol{g}$. To put the sampled coefficients on a common Gaussian scale, choose an invertible factor $\boldsymbol{B}\boldsymbol{B}^T = \boldsymbol{V}_g$ and set $\boldsymbol{g}(\boldsymbol{z}) = \boldsymbol{m}_g + \boldsymbol{B}\boldsymbol{z}$. Then Equation (20) gives $\mu(\boldsymbol{g}(\boldsymbol{z})) = m_\gamma + \boldsymbol{\lambda}^T \boldsymbol{z}$, where $\boldsymbol{\lambda} = \boldsymbol{B}^{-1}\boldsymbol{v}$ is computed by a linear solve. Under the ordinary posterior, $\boldsymbol{z}$ has a standard normal distribution. The shape restriction then enters through the survival weight in Equation (14).

When computing s^2, subtracting two nearly equal terms in Equation (20) can lose precision. The equivalent formula $s^2 = (\boldsymbol{d}^T \boldsymbol{V}^{-1} \boldsymbol{d})^{-1}$, where $\boldsymbol{d}$ is the column of $\boldsymbol{A}^{-1}$ corresponding to γ, avoids this subtraction. The quadratic form is the γ diagonal entry of the transformed precision matrix and can be evaluated using the factored posterior precision without forming $\boldsymbol{V}^{-1}$ explicitly.

The ordinary Gaussian posterior can itself be prepared using matrix factorizations and linear solves. For a general noise covariance $\boldsymbol{\Sigma}$, solves with its Cholesky factor transform the observation

errors to have identity covariance. We apply the same transformation to the design and to the response centered at $\boldsymbol{X}\boldsymbol{m}_0$, then scale the design columns by the prior standard deviations. In these coordinates, the prior precision is the identity. Stacking the transformed design above an identity matrix and applying a QR factorization therefore gives a triangular factor of the posterior precision. Linear solves supply the posterior mean and covariance factors, after which the full transformation $\boldsymbol{A}\boldsymbol{V}\boldsymbol{A}^T$ preserves the dependence introduced by the coefficient map.

Algorithm 1 separates the repeated marginal-density evaluation from reconstruction after sampling. Both stages use the same threshold rule, tolerance, and precision policy, so they describe the same conservative conditioning event. A gradient-based sampler also needs a vector to guide its proposals, called the force below. We obtain it from the envelope derivative in Section B.3, which differentiates the maximized objective without differentiating the numerical search for its maximizing location.

Algorithm 1 Sign-constrained marginal sampling and reconstruction. For a joint event satisfying Proposition 1, use its common coefficient map and replace τ_r by τ_A^ε throughout, including the upper threshold used in both stages.

Require: Gaussian model, event $\mathcal{C}_r$ or $\mathcal{C}_A^\varepsilon$, admissible map, interval I
Require: Absolute tolerance $\delta > 0$ and precision budget
Form the ordinary posterior and apply the invertible shape-specific map
Prepare $\boldsymbol{m}_g, \boldsymbol{B}, m_\gamma, \boldsymbol{\lambda}, s$
function MARGINAL EVALUATION($\boldsymbol{z}$)
    Set $\boldsymbol{g} = \boldsymbol{m}_g + \boldsymbol{B}\boldsymbol{z}$
    Set $\mu = m_\gamma + \boldsymbol{\lambda}^T\boldsymbol{z}$
    Bound $\tau_r(\boldsymbol{g})$ between ℓ and u with $u - \ell \le \delta$, or report failure
    Set $\widetilde{\tau}_r = u$ and $\widetilde{\alpha} = (\widetilde{\tau}_r - \mu)/s$
    Return $-\|\boldsymbol{z}\|^2/2 + \log \Phi^{\mathrm{c}}(\widetilde{\alpha})$ and the envelope force
end function
Use the marginal sampler to retain draws of $\boldsymbol{z}$
for each retained $\boldsymbol{z}$ **do**
    Set $\boldsymbol{g} = \boldsymbol{m}_g + \boldsymbol{B}\boldsymbol{z}$ and $\mu = m_\gamma + \boldsymbol{\lambda}^T\boldsymbol{z}$
    Evaluate the same upper-threshold rule and set $\widetilde{\alpha} = (\widetilde{\tau}_r - \mu)/s$
    Draw $U \sim \mathrm{Uniform}(0, 1)$ independently
    Compute $\eta = -\Phi^{-1}(U\Phi^{\mathrm{c}}(\widetilde{\alpha}))$ using log probabilities
    Set $\gamma = \widetilde{\tau}_r + s(\eta - \widetilde{\alpha})$ and invert the coefficient map
    Evaluate the reconstructed curve and requested functionals
end for

B.2 Conservative threshold evaluation

The threshold is a maximum over the entire constraint interval. Evaluating its objective at finitely many locations gives a lower bound, while an upper bound must also control values between those locations. The calculation therefore maintains both bounds and refines them until their difference is within the requested tolerance.

Bounds used for conditioning. The routine takes the remaining coefficients in their declared order, the shape case and basis geometry, a positive absolute tolerance δ in coefficient units, and an arithmetic precision limit. A successful evaluation returns

$$\ell \le \tau_r(\boldsymbol{g}) \le u, \qquad u - \ell \le \delta,$$

together with an approximate maximizing location for the envelope force. Choosing $\widetilde{\tau}_r = u$ then ensures $0 \le \widetilde{\tau}_r - \tau_r \le \delta$. The resulting lower bound on γ can be slightly too large, but cannot admit a curve that violates the constraint. If the requested enclosure cannot be obtained, the routine reports failure.

Locating and evaluating extrema. For each individual constraint, the maximum occurs at an endpoint or an interior stationary location. Corollaries 1 to 3 express the interior candidates as roots of a polynomial in the cosine coordinate $w \in J$. Real-root isolation finds disjoint intervals containing every distinct real root and establishes that none have been omitted (Johansson, 2017). The represented coefficients are treated as exact rationals, allowing repeated factors to be removed before isolation while retaining roots of even multiplicity. This arithmetic also detects reduced degree without discarding small nonzero coefficients. A constant stationary polynomial requires only endpoint evaluations, including when it is identically zero.

Interval arithmetic then encloses the objective at the endpoints and over each root interval. Rounding is directed outward so that these enclosures include arithmetic error. The largest candidate upper bound gives u, while valid lower bounds come from endpoints and roots known to belong to J. Refining the candidate enclosures reduces the gap between these bounds until it is at most δ.

Roots near the boundary of J need additional care. A root interval known to lie outside J is discarded. One crossing the boundary remains an upper-bound candidate, but supplies a lower bound only after its root is shown to belong to J. The calculation can still stop without resolving that location if the overall gap is already within δ. For quotient objectives, division also requires a strictly positive interval lower bound for the denominator.

Evaluation for the individual constraints. For positivity, isolate the roots of $P_{0,\boldsymbol{g}}$ and evaluate the objective $-f_{\boldsymbol{g}}$ over the corresponding physical locations. Interval evaluation of this trigonometric expression encloses the candidate values.

For monotonicity, the objective is $-P_{\boldsymbol{g}}/D$, with stationary polynomial

$$\frac{\mathrm{d}P_{\boldsymbol{g}}}{\mathrm{d}w} D - P_{\boldsymbol{g}} \frac{\mathrm{d}D}{\mathrm{d}w}.$$

The polynomials in Equation (9) use every spectral coefficient, with $g_2 = 0$ and the intercept excluded, and satisfy $D \ge 1$ on J. The lidar implementation first bounds the equivalent trigonometric expression, retaining all harmonics, and uses root isolation as a fallback.

For convexity, the polynomial objective $Q_{\boldsymbol{g}}$ can be bounded before finding its stationary points. Map J affinely to $[0, 1]$ and express the polynomial in the Bernstein basis. These basis functions are nonnegative and sum to one, so every polynomial value is a weighted average of the coefficients and lies between their smallest and largest values. Ball arithmetic, which represents each quantity by a midpoint and an error radius, encloses the change of basis and the cosine endpoints.

De Casteljau subdivision rewrites the polynomial on smaller intervals, with outward rounding preserving the bounds. Repeatedly subdividing the interval with the largest upper bound refines u, while endpoint and midpoint evaluations improve ℓ. If the gap still exceeds δ when the subdivision budget is reached, isolate the roots of the exact rational derivative of $Q_{\boldsymbol{g}}$ and increase arithmetic precision as needed.

Joint restrictions. For a signed joint event, use a common coefficient map satisfying Proposition 1. Every selected order then shares the same direction h and remaining coefficients $\boldsymbol{g}$, so its quotient objective must be formed from Equation (13) under this map. Subdivide I and bound each quotient on every subinterval, verifying a positive lower bound for its signed denominator before division. Evaluations at points in I supply lower bounds.

If $[\ell_r, u_r]$ encloses the maximum for order $r \in A$, then

$$\max_{r \in A} \ell_r \leq \tau_A^{\varepsilon}(\boldsymbol{g}) \leq \max_{r \in A} u_r.$$

The width of this joint enclosure is at most $\max_{r \in A}(u_r - \ell_r)$, so requiring each width to be at most δ suffices. The reductions in Corollary 4 and Table 4 replace some continuum searches by endpoint evaluations. Enclose each retained maximum and endpoint quotient with outward rounding, then take the maxima of their lower and upper bounds. The same width guarantee applies. The positive–convex and negative–concave pairs retain the continuum value threshold M_0 alongside the curvature threshold M_2.

Precision and error control. These guarantees concern the numerical coefficients and geometry supplied to the routine, treated as exact inputs. They exclude errors from the preceding Gaussian preparation. Applying Proposition 2 requires the threshold-error bound to hold almost everywhere under the ordinary Gaussian law.

Fixing the precision policy and the rules for choosing among candidate locations makes repeated evaluation deterministic. Nonfinite inputs and enclosures unresolved at the precision limit are reported as failures. The Gaussian survival weight is positive at every finite coefficient vector, so assigning zero density to an unresolved evaluation would impose an additional restriction. Such failures therefore cannot be silently treated as rejected coefficient values.

B.3 Threshold derivatives

Gradient-based sampling requires the derivative of the log marginal density with respect to the sampled coefficients. The shape restriction enters through the Gaussian survival weight in Equation (14), so this calculation depends on how the threshold changes with the coefficients. The maximum representation of the threshold gives these derivatives from its maximizing locations, without differentiating the numerical search used to find them.

Slopes at maximizing locations. At a fixed x, the threshold objective is linear in $\boldsymbol{g}$. Under Theorem 1, write

$$\tau_r(\boldsymbol{g}) = \max_{x \in I} \boldsymbol{\beta}_r(x)^T \boldsymbol{g},$$

where $\boldsymbol{\beta}_r(x)$ contains the coefficient multipliers, ordered as the entries of $\boldsymbol{g}$. The positive denominator and compact interval make this vector continuous and bounded in x. Consequently, τ_r is convex, positively homogeneous, and Lipschitz, with Lipschitz constant at most $\max_{x \in I} \|\boldsymbol{\beta}_r(x)\|$.

Let $\mathcal{A}_r(\boldsymbol{g})$ denote the locations attaining the maximum, which we call active locations. For a perturbation in a direction $\boldsymbol{b} \in \mathbb{R}^m$, Danskin's theorem (Bertsekas, 2010, Exercise 6.7) gives

$$\frac{\mathrm{d}}{\mathrm{d}u} \tau_r(\boldsymbol{g} + u\boldsymbol{b}) \bigg|_{u=0+} = \max_{x \in \mathcal{A}_r(\boldsymbol{g})} \boldsymbol{\beta}_r(x)^T \boldsymbol{b}.$$

The theorem applies because I is compact and the coefficient gradients are continuous. Its formula compares the slopes at the currently maximizing locations and takes the largest slope in the chosen direction.

The subdifferential consists of all convex combinations of the active vectors $\boldsymbol{\beta}_r(x)$. A gradient exists exactly when all these vectors agree, in which case their common value is the column gradient. Thus a unique maximizing location suffices for differentiability. Several locations also permit a gradient if they give the same vector, while different active vectors produce a kink.

Individual and joint constraints. For the individual constraints, the coefficient vectors follow directly from the objectives in Corollaries 1 to 3. With $w(x) = \cos(\omega_1(x - c + L))$, their spectral components are

$$\begin{aligned}
\beta_{0,j}(x) &= -\phi_{j,L}(x), && 1 \le j \le m, \\
\beta_{1,j}(x) &= -jT_j(w(x))/D(w(x)), && 1 \le j \le m, \quad j \ne 2, \\
\beta_{2,j}(x) &= j^2 U_{j-1}(w(x)), && 2 \le j \le m.
\end{aligned}$$

The intercept components in the monotonicity and convexity coordinates are zero. Where the threshold is differentiable, these formulas evaluate its gradient at any maximizing location.

For the joint threshold in Proposition 1, form the vectors from the common coefficient map so that $\eta_r(x; \boldsymbol{g}) = \boldsymbol{\beta}_r(x)^T \boldsymbol{g}$. An active candidate is now a pair (r, x) attaining the joint threshold. The directional derivative is the largest $\boldsymbol{\beta}_r(x)^T \boldsymbol{b}$ among these pairs, and a gradient exists exactly when their vectors agree. This includes ties between different derivative orders. The reductions in Corollary 4 preserve the threshold, so the same calculation can use the retained continuum objectives and endpoint quotients.

Gradient of the marginal log density. Let $q_r(\boldsymbol{z})$ be the exact conditioned density in whitened coordinates from Equation (14). Define the standardized threshold by

$$\alpha_r(\boldsymbol{z}) = \frac{\tau_r(\boldsymbol{g}(\boldsymbol{z})) - \mu(\boldsymbol{g}(\boldsymbol{z}))}{s},$$

and let φ_1 denote the standard normal density. At a point where the active coefficient vectors agree, choose any maximizing location $x_\star$. The map $\boldsymbol{g}(\boldsymbol{z}) = \boldsymbol{m}_g + \boldsymbol{B}\boldsymbol{z}$ transforms the threshold gradient to $\boldsymbol{B}^T \boldsymbol{\beta}_r(x_\star)$, while the conditional mean has gradient $\boldsymbol{\lambda}$. The chain rule therefore gives the column gradients

$$\begin{aligned}
\frac{\partial \alpha_r}{\partial \boldsymbol{z}} &= \frac{\boldsymbol{B}^T \boldsymbol{\beta}_r(x_\star) - \boldsymbol{\lambda}}{s}, \\
\frac{\partial \log q_r}{\partial \boldsymbol{z}} &= -\boldsymbol{z} - \frac{\varphi_1(\alpha_r)}{\Phi^{\mathrm{c}}(\alpha_r)} \frac{\partial \alpha_r}{\partial \boldsymbol{z}}.
\end{aligned} \tag{21}$$

The term $-\boldsymbol{z}$ comes from the standard normal density, while the second term differentiates the log survival probability. Its weight $\varphi_1(\alpha_r)/\Phi^{\mathrm{c}}(\alpha_r)$ is the inverse Mills ratio. The gradient of α_r accounts for changes in both the threshold and the conditional mean of γ.

The numerical force. The numerical adapter selects an approximate maximizing location deterministically, evaluates $\boldsymbol{\beta}_r$ there, and uses $\widetilde{\alpha} = (\widetilde{\tau}_r - \mu)/s$ in the normal-tail factor. Substituting these quantities in the envelope formula gives the force used by the sampler.

The threshold tolerance controls objective values, but locations with almost equal values can have substantially different coefficient slopes. A narrow enclosure of the maximum therefore does not guarantee an accurate force near a change of maximizing location. The numerical log density uses the upper threshold, so the returned envelope force need not be its exact gradient. Section C explains how this force is used in sampling.

B.4 Stable tail evaluation and reconstruction

When the threshold lies far above the scalar conditional mean, its Gaussian survival probability can be extremely small. Computing $1 - \Phi(\alpha)$ can then lose precision or return zero, even though the probability is positive. We instead evaluate $\log \Phi^{\mathrm{c}}(\alpha)$ directly as $\log \Phi(-\alpha)$ using a normal log-tail routine. The inverse Mills ratio in Equation (21) is then $\exp(\log \varphi_1(\alpha) - \log \Phi^{\mathrm{c}}(\alpha))$.

For reconstruction, draw $U \sim \mathrm{Uniform}(0, 1)$ independently and use a normal quantile routine that accepts log probabilities to solve

$$\log \Phi(-\eta) = \log U + \log \Phi^{\mathrm{c}}(\widetilde{\alpha}).$$

In exact arithmetic this gives $\eta > \widetilde{\alpha}$ for $0 < U < 1$, and $\gamma = \widetilde{\tau}_r + s(\eta - \widetilde{\alpha})$ equals the truncated-normal draw in Equation (7) with the upper threshold substituted. Expressing the draw relative to the threshold is useful when the conditional mean and a large positive normal quantile would nearly cancel. A nonfinite quantile or a computed $\eta < \widetilde{\alpha}$ is a reconstruction failure. These arithmetic safeguards do not extend Proposition 2 to rounding in Gaussian preparation or tail evaluation, since that bound assumes exact Gaussian and tail operations.

C Implementation in Stan

For monotonicity and convexity, the Stan programs implement Algorithm 1 using the No-U-Turn sampler (NUTS) (Hoffman and Gelman, 2014; Carpenter et al., 2017). The sampled parameter is the unconstrained vector $\boldsymbol{z} \in \mathbb{R}^m$. Gaussian preparation supplies $\boldsymbol{m}_g, \boldsymbol{B}, m_\gamma, \boldsymbol{\lambda}$, and $s > 0$ as data, together with the basis geometry, constraint interval, and threshold tolerance. The threshold routine also has a specified precision budget. The rank and padding satisfy the conditions of Corollaries 2 and 3. Preparation allows a general positive-definite noise covariance. The convexity program uses a basis centered at zero, so a model centered at c is represented by subtracting c from both observation and constraint coordinates.

At each target evaluation, the program forms $\boldsymbol{g} = \boldsymbol{m}_g + \boldsymbol{B}\boldsymbol{z}$ and $\mu = m_\gamma + \boldsymbol{\lambda}^T \boldsymbol{z}$, then evaluates the upper threshold $\widetilde{\tau}_r(\boldsymbol{g})$. In Listing 1, `threshold_upper` denotes this value, `conditional_sd` denotes s, and `alpha` denotes $\widetilde{\alpha} = (\widetilde{\tau}_r(\boldsymbol{g}) - \mu)/s$. The Gaussian inputs already incorporate the observations, so the model block adds only the log survival weight `log_survival`$= \log \Phi^{\mathrm{c}}(\widetilde{\alpha})$ to the standard normal log density of $\boldsymbol{z}$.

External C++ functions supply the threshold and normal-tail operations. The threshold function follows Section B.2 and attaches the approximate envelope derivative from Section B.3 to its returned upper value through a custom automatic differentiation adapter. Stan can therefore propagate the force through the linear coefficient map without differentiating the root search or interval subdivision. The log-survival adapter uses the negative inverse Mills ratio, and the quantile routine accepts a log probability, as described in Section B.4.

For each retained $\boldsymbol{z}$, the generated-quantities block draws γ from its truncated conditional Gaussian law using an independent uniform variate and the same upper threshold as the model block. Inverting the coefficient map then gives a joint coefficient draw. In particular, the monotonicity program restores $a_2 = -\gamma$ and $a_4 = g_4 - \gamma/4$, while the convexity program restores $a_1 = -\gamma$. The other coefficients follow Table 2. The finite-value and quantile checks in Section B.4 apply to this reconstruction.

Extending these programs to a joint restriction requires Gaussian inputs and reconstruction under the common map in Corollary 4. The external threshold routine must enclose τ_A^ε and supply the envelope force from its active orders and locations, as described in Sections B.2 and B.3. With these inputs, the scalar sampling and reconstruction blocks in Listing 1 apply unchanged.

```
model {
  target += -0.5 * dot_self(z) + log_survival;
}
generated quantities {
  real eta = -normal_log_quantile(
    log_survival + log(uniform_rng(0, 1)));
  real gamma =
    threshold_upper + conditional_sd * (eta - alpha);
}
```

Listing 1: Marginal sampling and conditional reconstruction in Stan. The excerpt assumes that the Gaussian transformations, upper threshold, and log survival weight have been evaluated. The generic routine `normal_log_quantile` returns $\Phi^{-1}(\exp(\ell))$ from a supplied log probability ℓ.

The returned log values determine the density used to weight NUTS states, whereas the supplied force determines the proposed trajectory. To justify this distinction, consider exact arithmetic with a fixed Euclidean metric, step size, and maximum tree depth after warm-up. Suppose the log values and force are finite, deterministic, measurable functions of position and the resulting density is normalizable. Leapfrog consists of position and momentum shears that preserve volume and invert when the step size changes sign. Consequently, momentum reversal makes the proposal reversible even when the force is not the log-density gradient (Neal, 2011, Section 5.5). With the recursive subtree checks and weighted state selection of NUTS (Betancourt, 2017, Appendices A.3–A.5), this preserves the density defined by the returned log values in the ideal construction without evaluation failures or divergence truncation. The depth limit makes each transition finite. This argument justifies using an approximate envelope force but gives no bound on finite-chain or floating-point error, and the force approximation can still affect sampling efficiency.

D Lidar application details

D.1 Computation

For the model in Section 5.1, computation subtracts the observed response mean and divides by the response range, applying the same change of scale to the Gaussian model. The global-threshold tolerance is 10^{-12} in these transformed coefficient units. An expectation propagation approximation at 129 derivative locations supplies the initial mean and dense proposal metric. The continuum threshold still determines every target evaluation, so these locations affect only initialization and the metric.

Four Stan chains each use 400 warm-up iterations and 2,000 retained draws, without thinning. During warm-up, the step size adapts to target acceptance 0.9 while the supplied metric is held fixed. The maximum tree depth is 10 during warm-up and 12 during retained sampling, which otherwise uses the resulting tuning.

MCMC diagnostics. We monitor the sampling coordinates and reconstructed coefficients, the latent response and concentration at the observed ranges, their adjacent midpoints, and five equally spaced locations, together with the total response increase, prior quadratic form, and log density. Across these 1,146 scalar quantities, the maximum rank-normalized split $\widehat{R}$ is 1.0034, and the minimum bulk and tail effective sample sizes are 2,776 and 2,479, respectively. None of the 8,000 retained iterations has a divergence or reaches the tree-depth limit, and the minimum energy Bayesian fraction of missing information (E-BFMI) across chains is 0.9637. Three divergences

occurred during warm-up, whose draws were discarded. The retained-draw diagnostics reveal no warning about mixing in the monitored quantities, while the predictive checks below assess agreement with the observation model.

D.2 Predictive checks

For each retained monotone posterior curve, define the standardized residuals $e_i(\boldsymbol{a}) = (y_i - f_{\boldsymbol{a}}(x(r_i)))/\sigma_i$. We generate one independent replicate dataset from

$$Y_i^{\text{rep}} \mid \boldsymbol{a} \overset{\text{ind}}{\sim} N(f_{\boldsymbol{a}}(x(r_i)), \sigma_i^2), \qquad e_i^{\text{rep}}(\boldsymbol{a}) = \frac{Y_i^{\text{rep}} - f_{\boldsymbol{a}}(x(r_i))}{\sigma_i}.$$

We compute the mean squared residual $n^{-1}\sum_i e_i(\boldsymbol{a})^2$, maximum $\max_i |e_i(\boldsymbol{a})|$, and absolute sample correlation between the paired vectors $(e_1, \ldots, e_{n-1})$ and $(e_2, \ldots, e_n)$, ordered by recorded distance. Here adjacency refers to consecutive released ranges, whose spacings vary. The two remaining discrepancies are the signed mean residuals over $r_i \leq 555$ meters and $r_i > 555$ meters, each averaged over the observations in its own half of the range. Applying the same definitions to e_i^{rep} gives the replicated discrepancies in Table 5.

Constrained posterior discrepancy	Lower tail	Upper tail
Mean squared standardized residual	0.5931	0.4069
Maximum absolute standardized residual	0.3914	0.6086
Absolute adjacent residual correlation	0.5359	0.4641
Mean standardized residual at $r \leq 555$ m	0.3526	0.6474
Mean standardized residual at $r > 555$ m	0.4235	0.5765

Table 5: Posterior predictive checks from 8,000 globally monotone posterior curves, with one independent Gaussian replicate dataset per curve. Lower and upper tails give the fractions of replicated discrepancies at or below and at or above the observed discrepancies for the same posterior curve, with the two range summaries split at the midpoint of 555 meters. The summaries condition on the fitted covariance and noise parameters and reuse the fitting data, so they are neither calibrated test probabilities nor out-of-sample predictive scores.

The upper-tail fractions in Table 5 range from 0.4069 to 0.6474, with no pronounced disagreement for these summaries under the specified observation model.

E Leaf application details

E.1 Data and computation

The sequence selected in Section 5.2 has sample identifier 1818. Of its 16 raw records, two are manually flagged and removed by the source study's quality filters. The third excluded record is the dark measurement, for which insufficient dark adaptation prevents estimation of dark respiration (Rogers et al., 2019b).

The marginal-likelihood optimization uses rank 32 and padding $L = 2$, with the resulting estimates carried into the rank-24 model in Equation (17). The constrained fit uses four Stan chains, each with 1,500 warm-up and 3,000 retained iterations, a dense metric, target acceptance 0.99, maximum tree depth 12, and threshold tolerance 10^{-8}. The unconstrained fit uses 20,000 independent Gaussian posterior draws.

MCMC diagnostics. We assess mixing across all 80 scalar outputs of the constrained fit, including the coefficient coordinates, reconstructed coefficients, log density, and quantities used in threshold evaluation and reconstruction. Across these outputs, the maximum rank-normalized split $\widehat{R}$ is 1.0007, and the minimum bulk and tail effective sample sizes are 5,616 and 7,866, respectively. None of the 12,000 retained iterations has a divergence or reaches the tree-depth limit, and the minimum energy Bayesian fraction of missing information (E-BFMI) across chains is 0.9642. These diagnostics reveal no warning about mixing in the monitored quantities. The predictive checks below separately assess agreement with the observation model.

E.2 Curvature inference

To complement the marginal-gain comparison in Figure 2, Figure 3 shows how rapidly marginal gain changes with irradiance. Concavity restricts the sign of this second derivative while allowing its magnitude to vary across the range. The pointwise intervals describe uncertainty about that local rate of change.

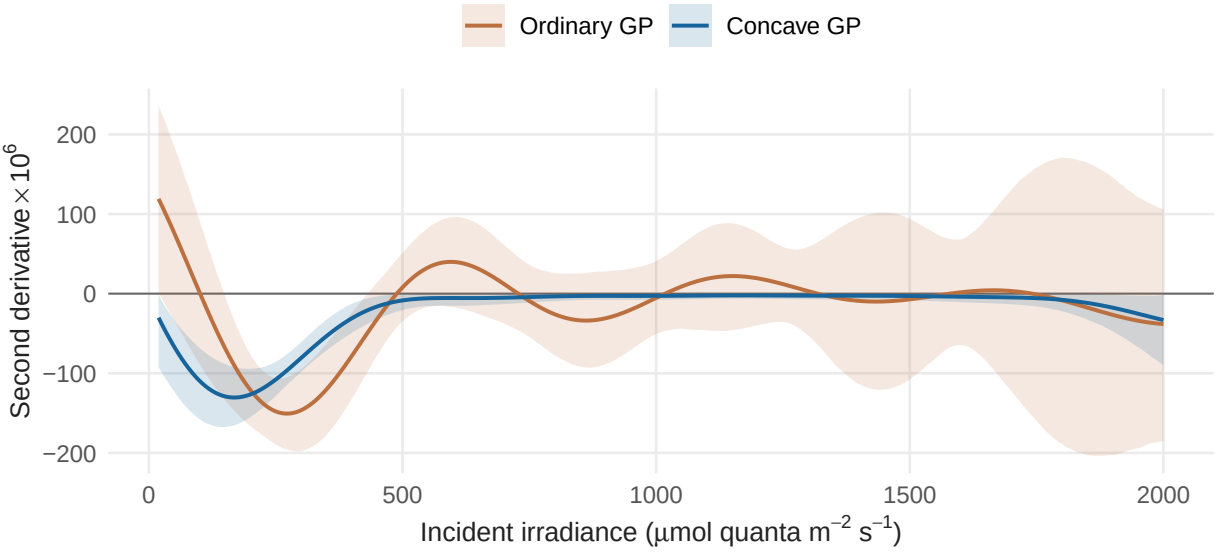


Figure 3: Second derivative of net assimilation with respect to irradiance for the selected leaf, multiplied by 10^6, with unscaled units of assimilation per squared irradiance. Orange and blue lines show ordinary Gaussian and concavity-conditioned posterior means and pointwise 95% equal-tail intervals under the same covariance and noise estimates. The intervals describe latent curvature, exclude measurement noise, and do not give simultaneous coverage, while the horizontal reference marks zero.

E.3 Predictive checks

For each posterior curve from either fit, we generate one independent replicate dataset at the observed irradiances using the Gaussian observation model in Equation (17). Define $r_i(A) = (y_i - A(q_i))/\sigma_\varepsilon$

and let π order the measurements by time. The two discrepancies are the maximum absolute standardized residual and the absolute sum of adjacent residual products,

$$D_{\max}(\boldsymbol{y}, A) = \max_i |r_i(A)|, \qquad D_{\text{seq}}(\boldsymbol{y}, A) = \left| \sum_{i=1}^{n-1} r_{\pi(i)}(A) r_{\pi(i+1)}(A) \right|.$$

For either discrepancy D, we average the indicator of a replicated value at least as large as the observed value over all curves from the corresponding fit. This upper-tail fraction estimates $P(D(\boldsymbol{Y}^{\text{rep}}, A) \geq D(\boldsymbol{y}, A) \mid \boldsymbol{y})$, with observed and replicated discrepancies evaluated using the same posterior curve.

The upper-tail fractions for $D_{\max}$ are 0.23 under concavity and 0.48 without the restriction, while D_{seq} gives 0.31 and 0.52, respectively. Neither comparison shows pronounced disagreement, although 13 records provide limited information about dependence or changes in measurement variance. Because irradiance decreases with time, the sequence discrepancy cannot distinguish temporal dependence from systematic response error across irradiance. Both fractions reuse the fitting data and therefore are not calibrated test probabilities.